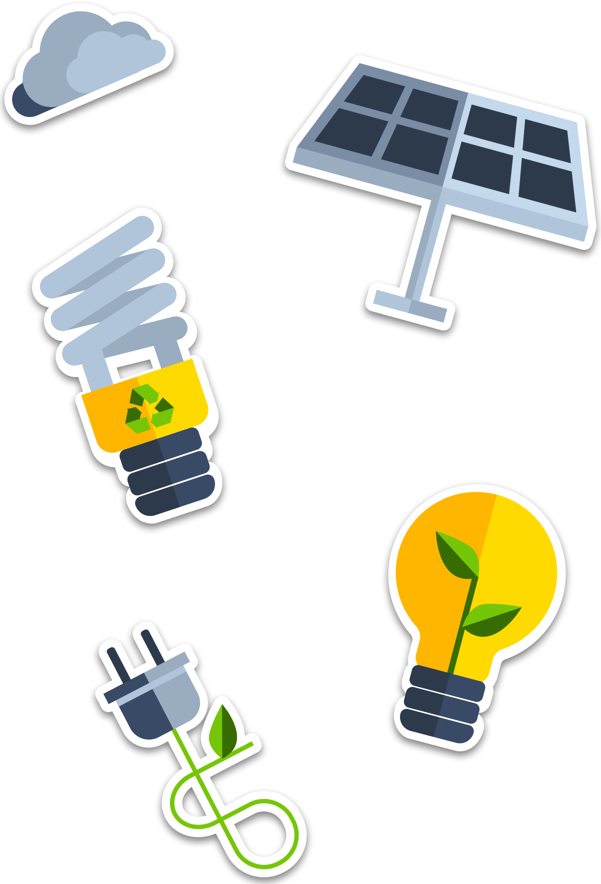

Star Harvester

Solar Energy Production

The world of solar energy

Our Team

Shu Xu

Semiconductor Technologist

Junjun Shen

Assembling/Welding Technologist

Bimla Danu

Theoretical Condensed Matter Physicist

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Natural Scientist / Bioinformatician

Outline

1. **INTRODUCTION** — Why solar energy matters
How we can improve the situation
2. **NATIONAL LEVEL** — Predicting (the development of)
Germany’s solar energy generation
3. **HOUSEHOLD LEVEL** — Predicting solar energy production
for a private household
4. **SUMMARY** — Status and outlook

Solar Energy Production

MOTIVATION

1. We want to convince potential customers (e.g., policy makers, families) of the immense potential of solar energy

2. We want to develop machine-learning models to accurately forecast daily solar energy generation in Germany

3. We want to provide advanced tools to the customers for efficient private solar energy production

Solar Energy - Germany

Annual total solar radiation
(averaged over 2001-2020)

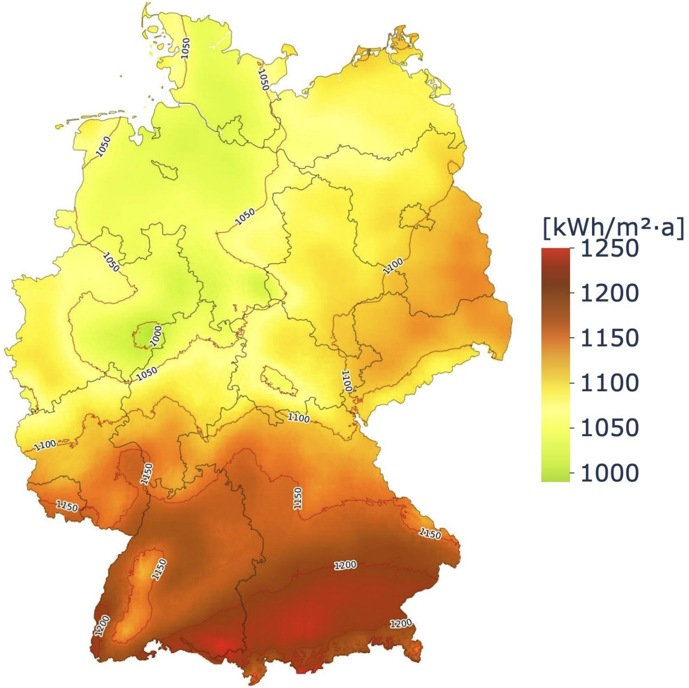
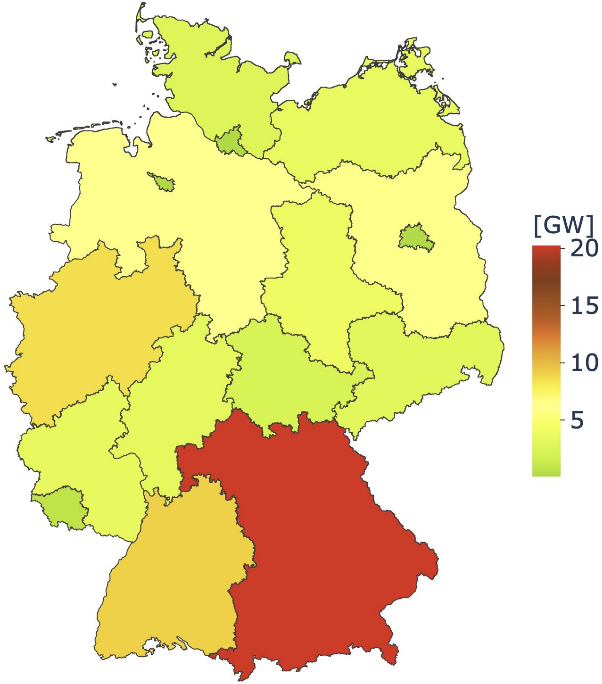


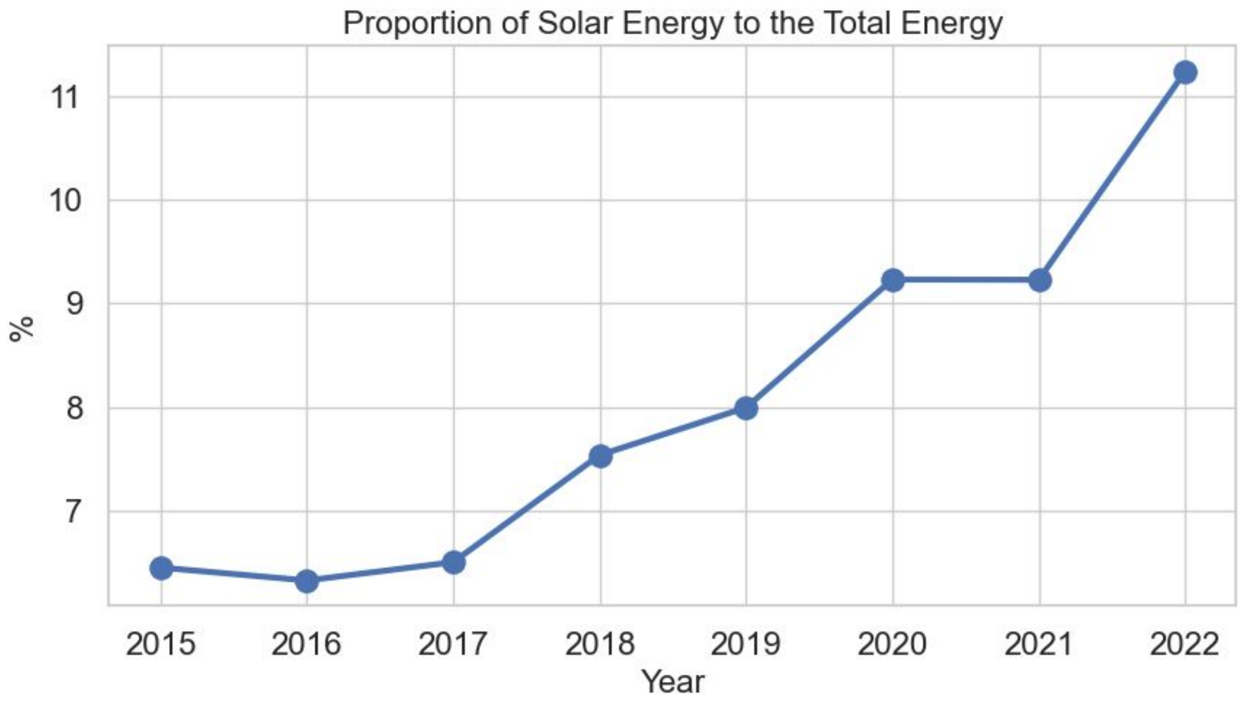
Image source: Fraunhofer ISE

Installed generation capacities
(up to June 2023)



Data source: Bundesnetzagentur

Solar Energy Generation - Germany



Average Energy Generation in Year 2022

Total Energy Generation - 493.37 [TWh]

Solar Energy Generation - 55.45 [TWh]

Main Data Sets

National Energy Production

- **Entso-e's Transparency Platform**

- 15-minute resolution
- Data since 2015
- ~300,000 observations

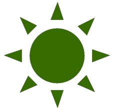
- Growing solar generation capacity

Household Solar Production

- **Private data from Görlitz**

- Daily resolution
- Data since 2005
- 6,630 observations

- Fixed capacity



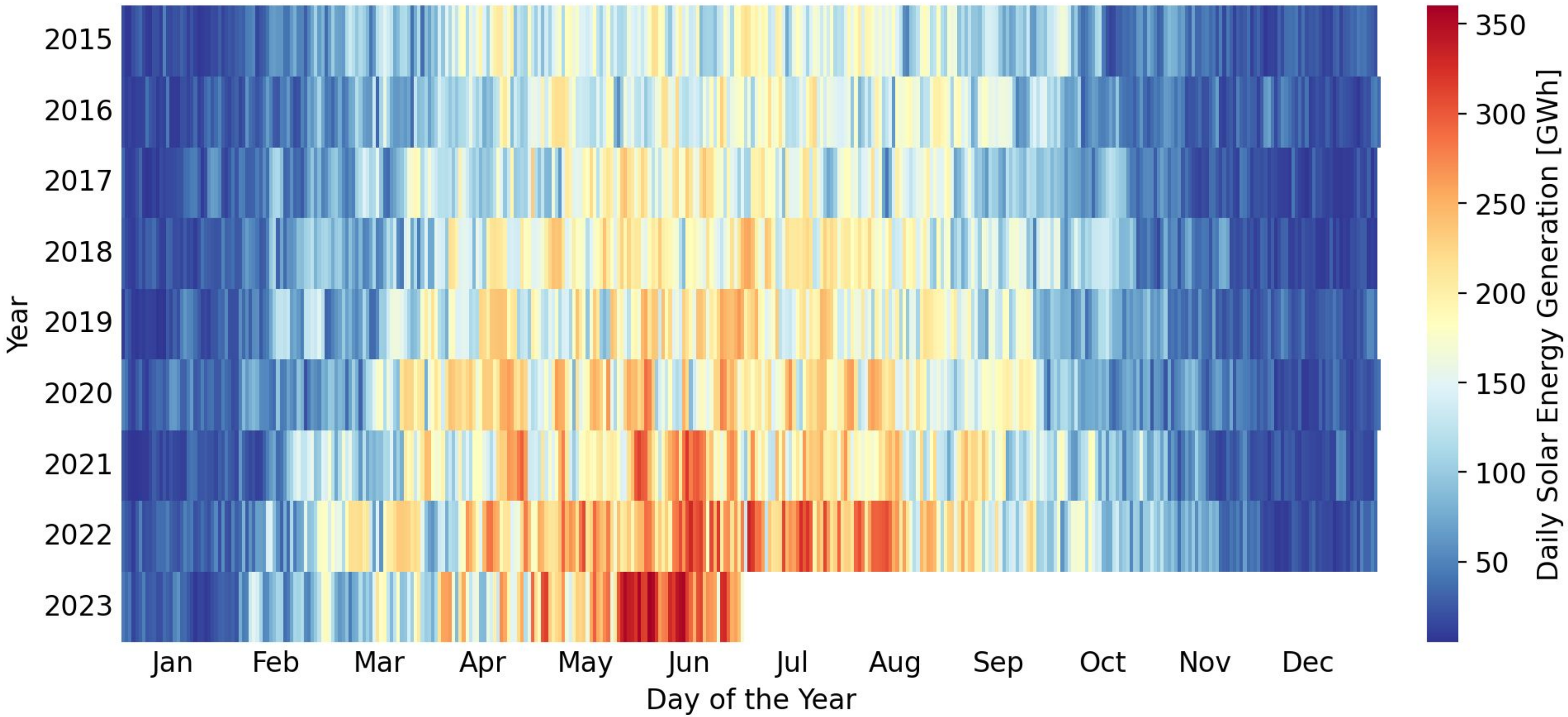
Weather Data

- **Deutscher Wetterdienst**

- Hourly or daily resolution
- Local or aggregated data

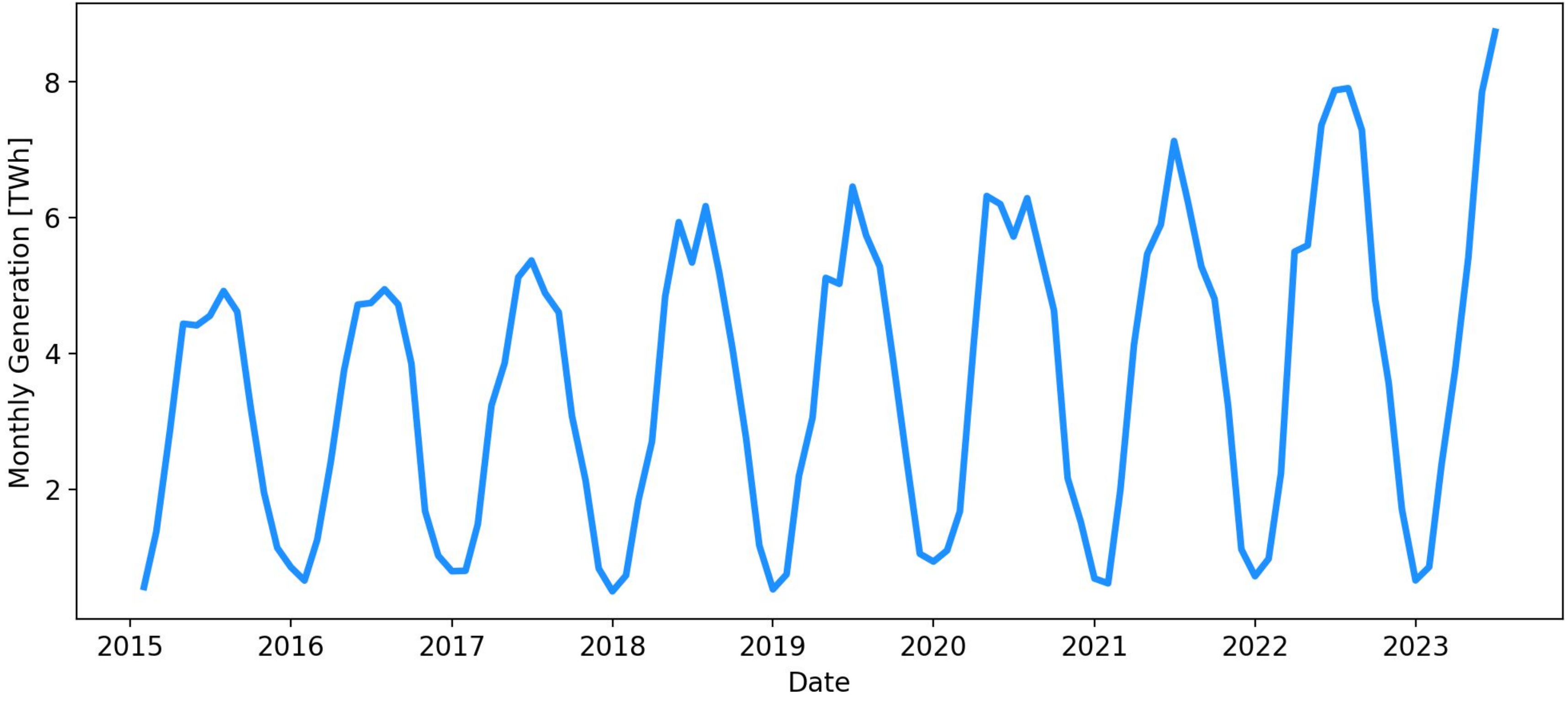
- Main features
  - Sunshine duration
  - Precipitation
  - Temperatures
  - Cloud coverage
  - Humidity
  - Windspeed

Germany's Solar Energy Generation



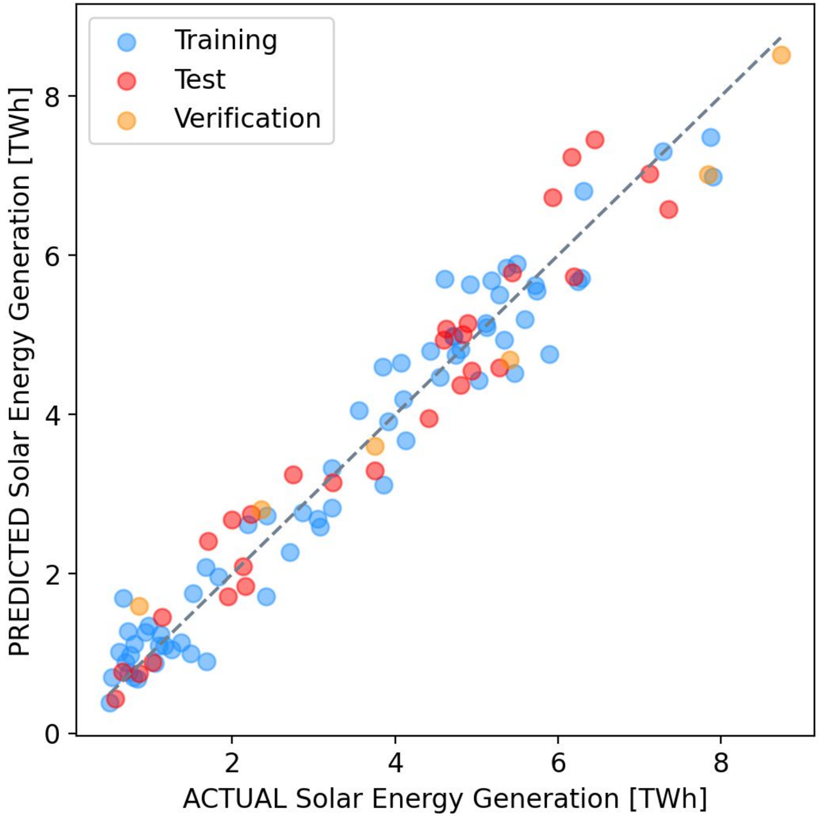
Data source: Entso-e's Transparency Platform

Monthly Solar Energy Generation



How to model it and make forecasts?

Monthly Solar Energy Generation



Linear model:

Monthly solar generation =
+ 1.522 * **average daily sunshine duration [hour]**
+ 0.046 * **total solar generation capacity [GW]**
+ 0.010 * average monthly precipitation [mm]
- 0.001 * average air temperature [°C]
- 3.068

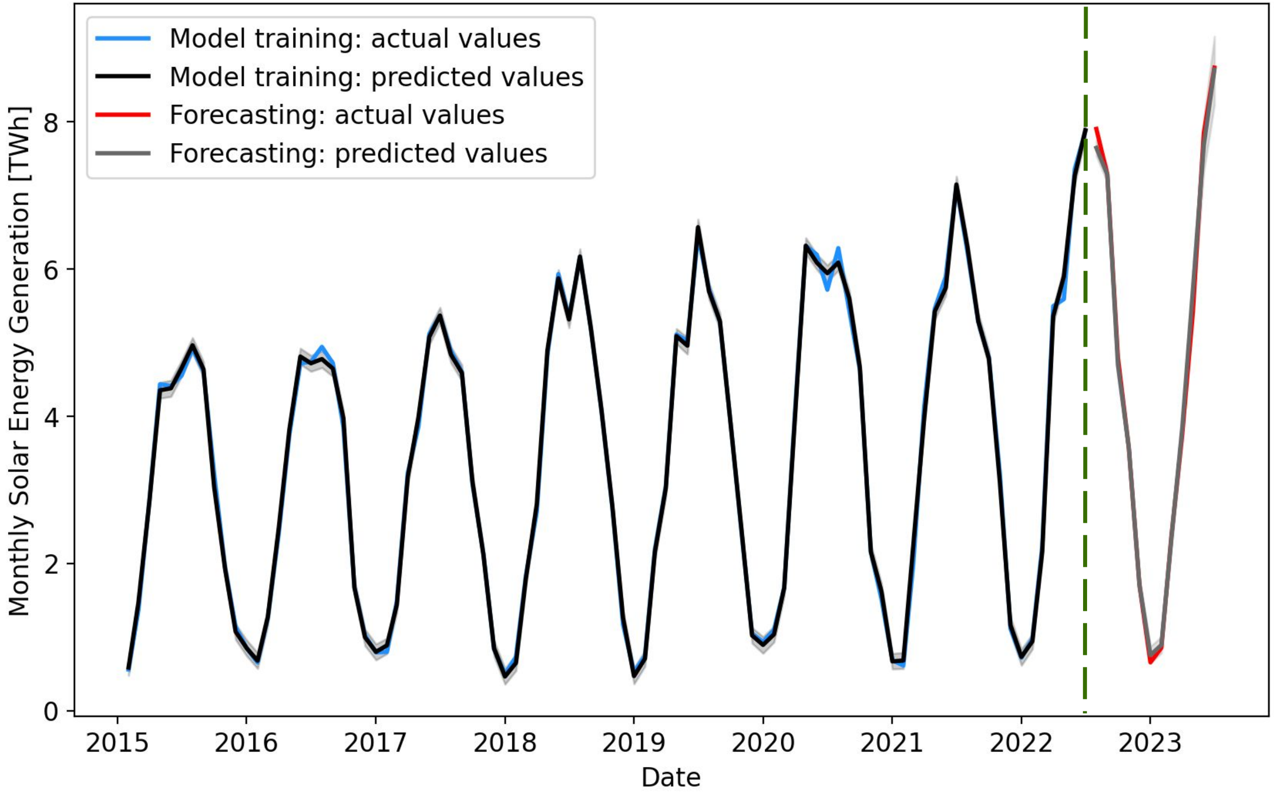
Average error of predictions:

Train: ~0.46 TWh
Test: ~0.50 TWh
Forecast: ~0.58 TWh

R^2 score: 0.94-0.96 (consistent)

Data sources: Deutscher Wetterdienst, Fraunhofer ISE, Bundesnetzagentur, Statistisches Bundesamt

Monthly Solar Energy Generation



Additional Input Variables

- Average air temperature
- Average sunshine duration
- Average precipitation
- Total solar generation capacity (updated monthly)

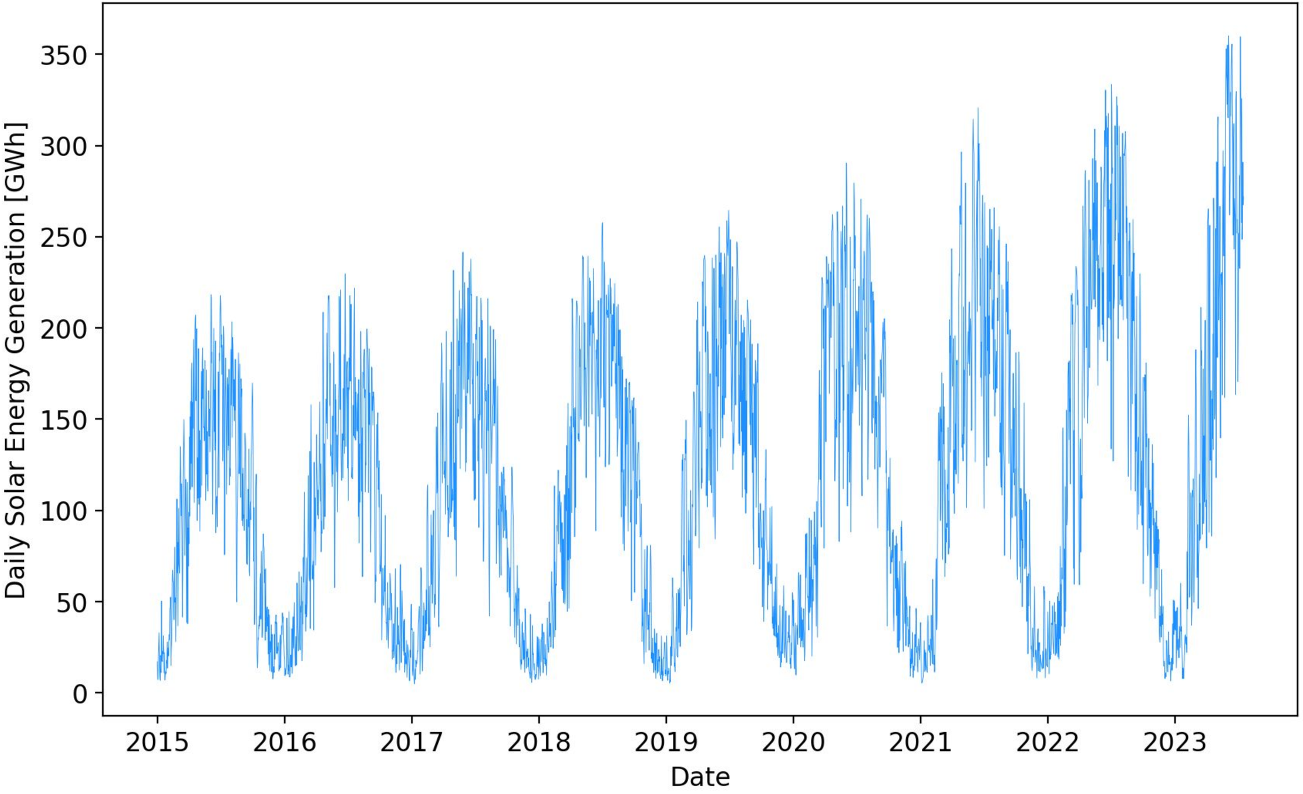
Average error of predictions:

Training: 0.086 TWh
Forecast: 0.155 TWh

R^2 score of predictions:

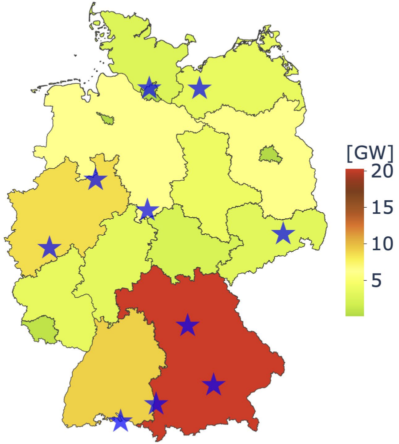
Training: 0.998
Forecast: 0.997

Daily Solar Energy Generation



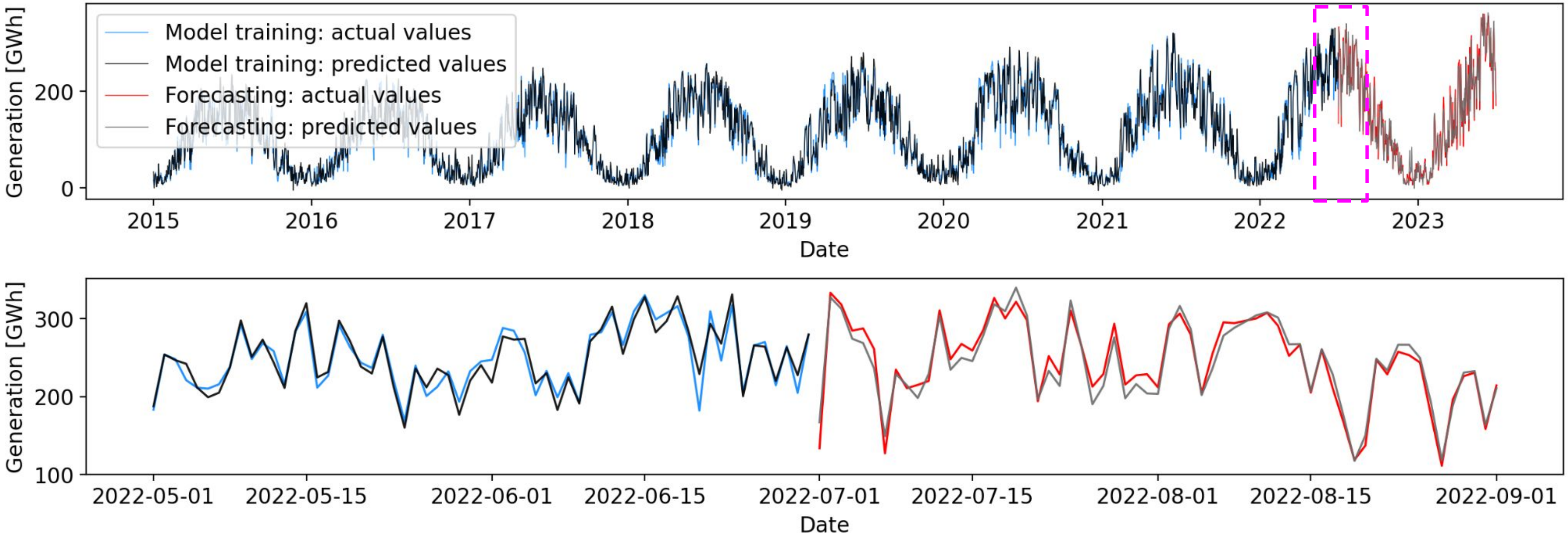
Input Weather Features
Average temperature [°C]
Norm temperature [°C]
Maximum temperature [°C]
Minimum temperature [°C]
Sunshine duration [hour]
Average wind speed [m/s]
Precipitation [mm]

Weather data averaged over 10 representative locations



Daily Solar Energy Generation

8 input variables: ***7 weather features + total solar generation capacity***



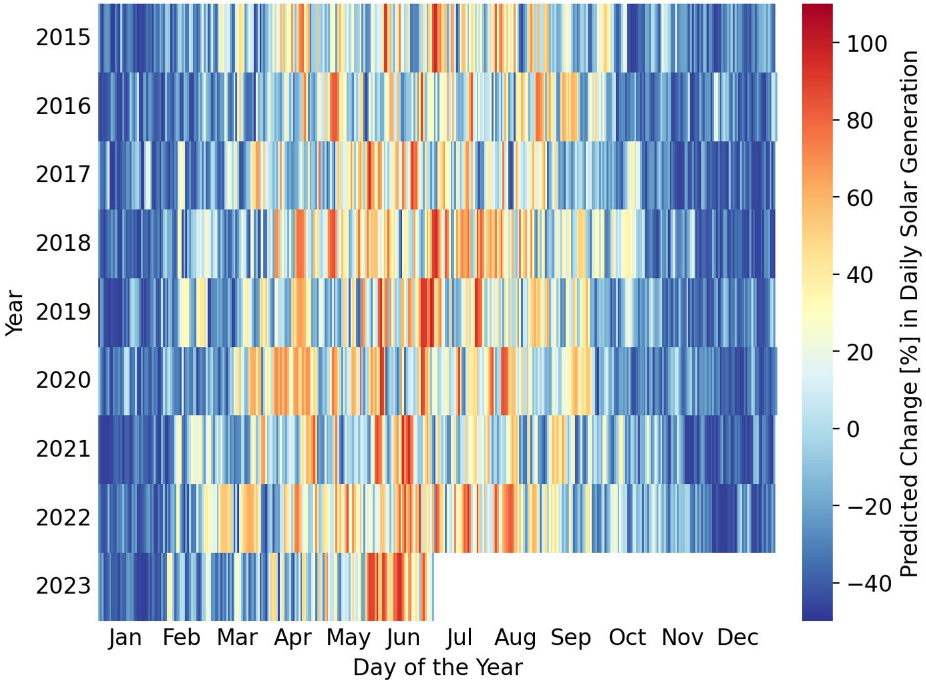
Average error of predictions:	
Training:	10.0 GWh
Forecast:	13.4 GWh

R^2 score of predictions:	
Training:	0.982
Forecast:	0.982

Daily Solar Energy Generation

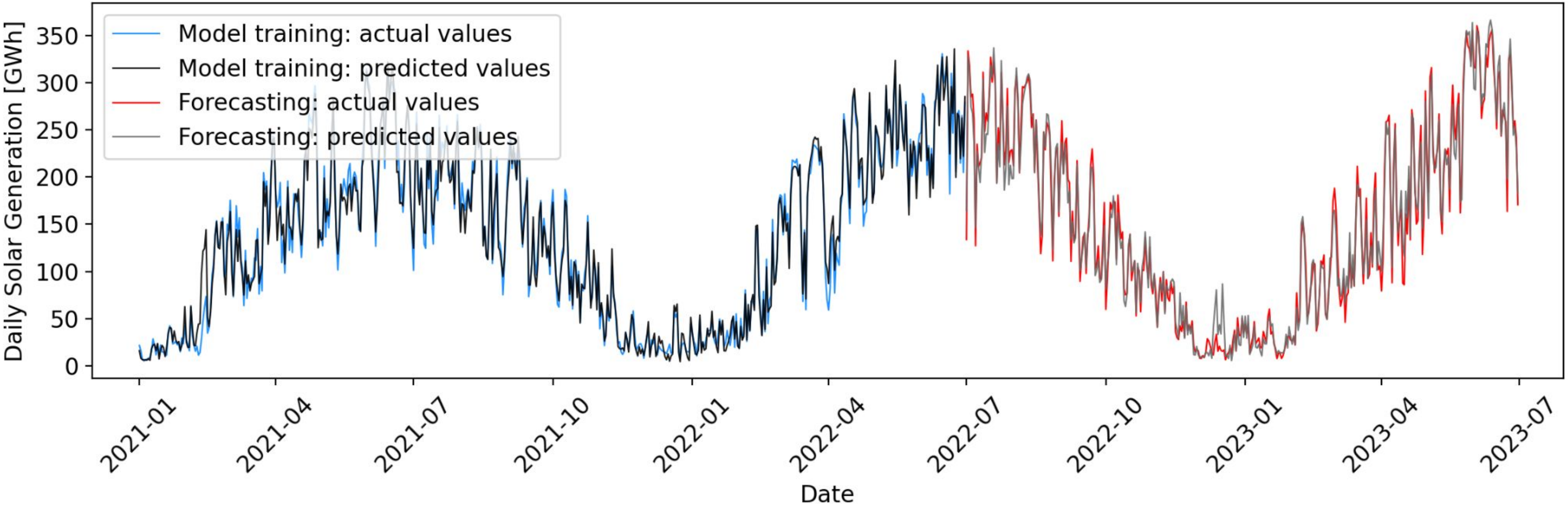
Input Variables	Mean Value	Coefficient [%]
Sunshine duration [hour]	**4.94**	**9.51**
Maximum temperature [°C]	14.84	1.05
Precipitation [mm]	1.97	-0.92
Average temperature [°C]	10.20	-0.75
Norm temperature [°C]	9.64	0.56
Average wind speed [m/s]	2.96	0.32
Minimum temperature [°C]	5.48	-0.21
Total solar generation capacity [GW]	46.69	-0.01

Predicted changes in daily solar generation caused by variable *'sunshine duration'*



Daily Solar Energy Generation

The only (additional) input variable: *'sunshine duration'*

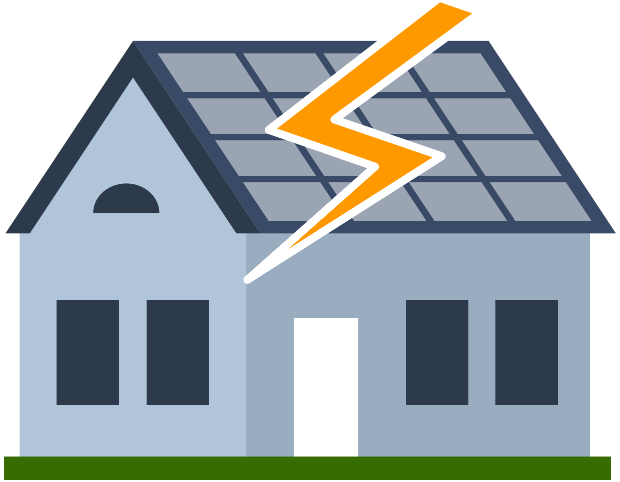


Average error of predictions:	
Training:	10.3 GWh
Forecast:	13.8 GWh

***R^2* score of predictions:**	
Training:	0.981
Forecast:	0.981

Forecasts using only one (additional) input variable

Our Product For Households



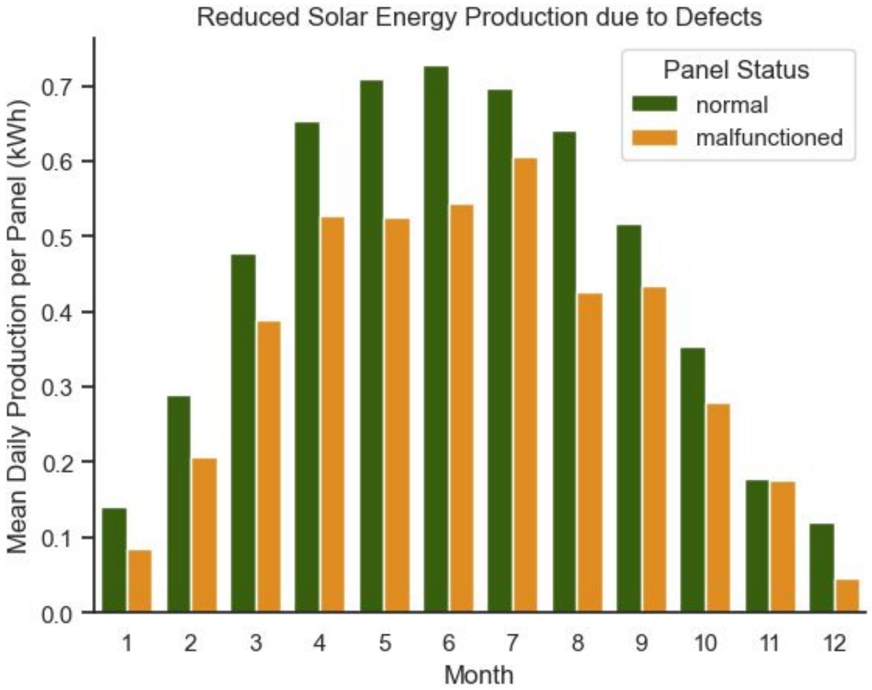
We predict the household solar energy production to

- provide insights in what affect the production to answer whether the investment is worthy or not.
- establish an alert system to detect reduced energy production in order to eliminate malfunctions.

Energy Production - Household

- Household in Goerlitz (GR)
- 18 years of continuous daily data
- 3 independent measurements (groups)
    - Identical panels
    - Identical installation conditions
- Group 1: free of malfunctioning
- Group 2 & 3: malfunctioned

Losses due to Malfunctioning



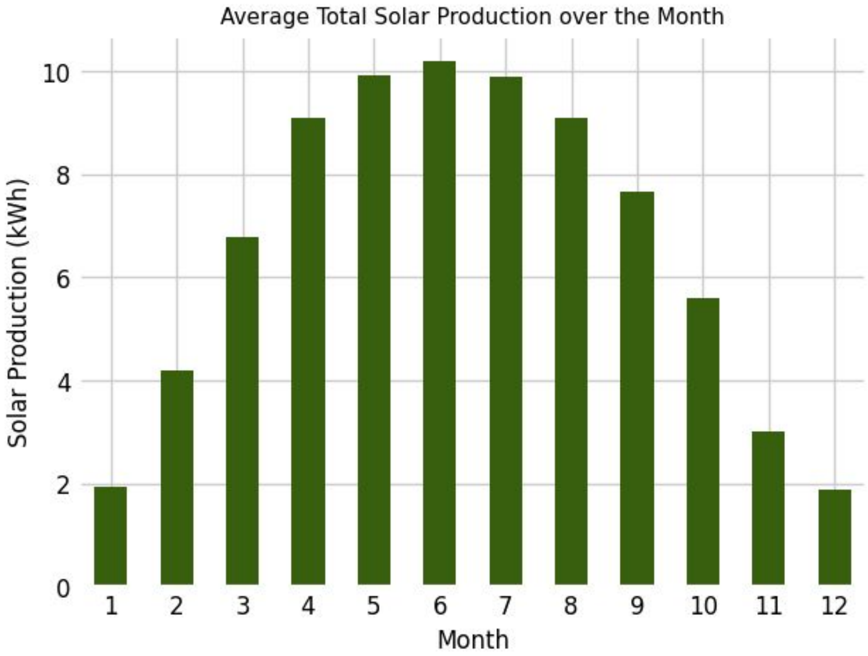
Figures of the solar panels

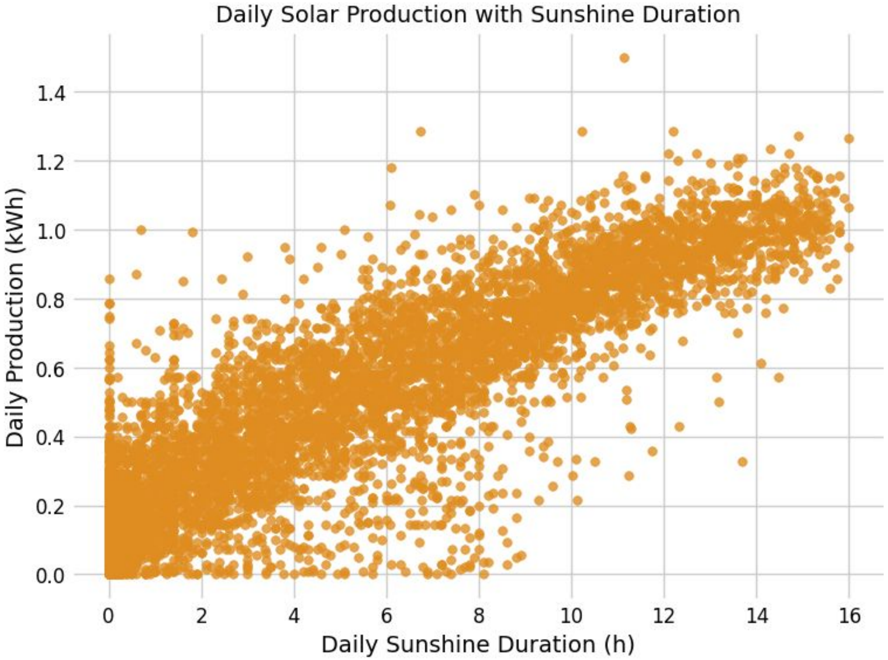
- Length of service
  - 6630 days
- In total malfunctioning
  - ~ 800 days
- Approximated total loss
  - 3654 kWh for 44 panels
  - 1993 € *

How to minimize the loss?

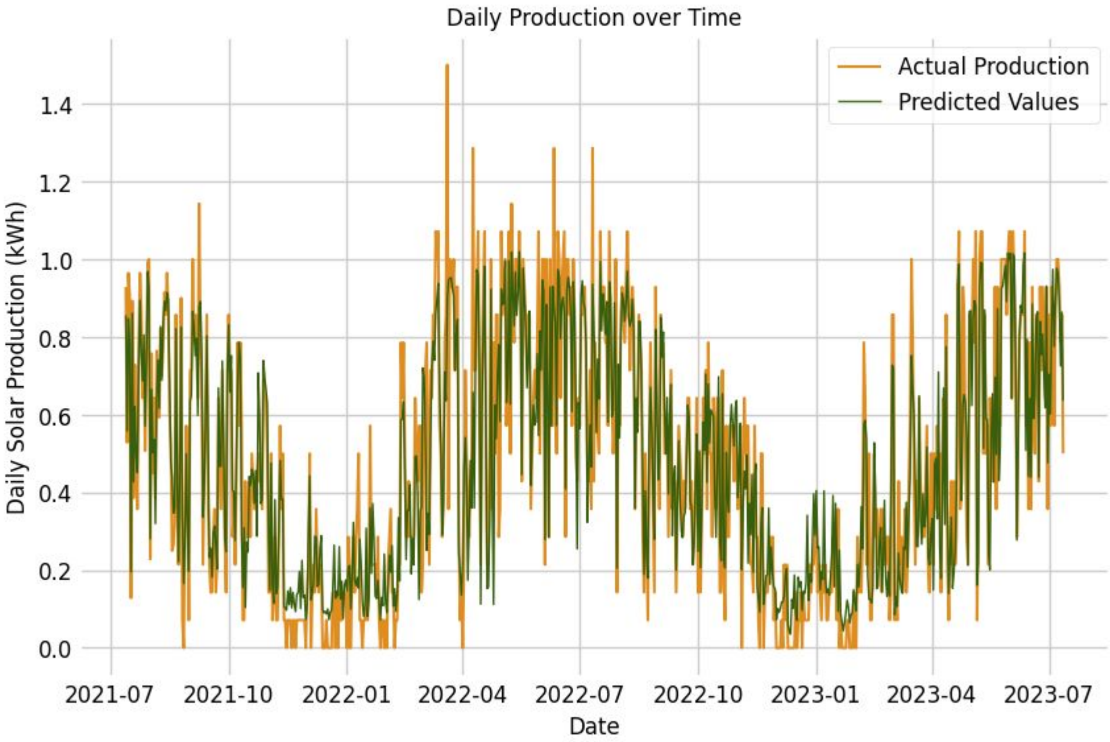
* *based on guaranteed price in 2005 acc. to EEG regulation from 2004-07-21*

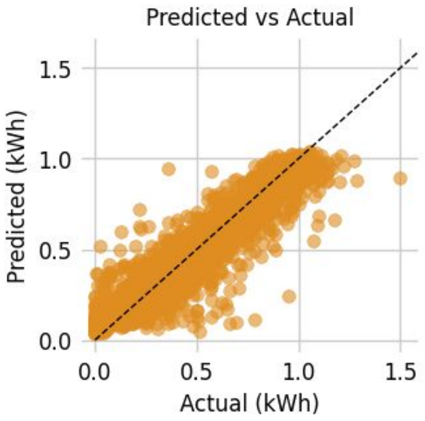
Model and Forecasting





Model and Forecasting





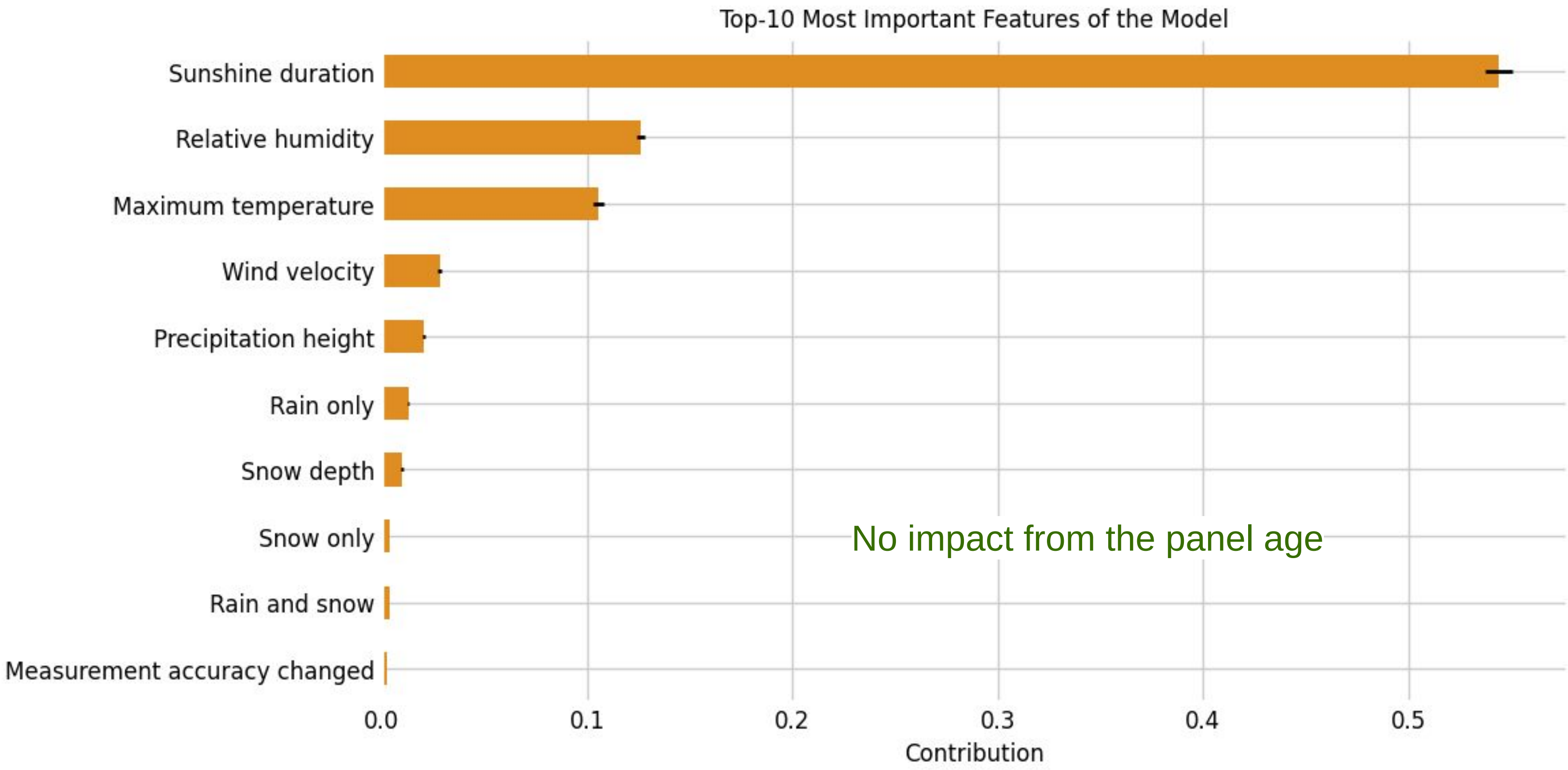
Regression model daily production

Model predictors

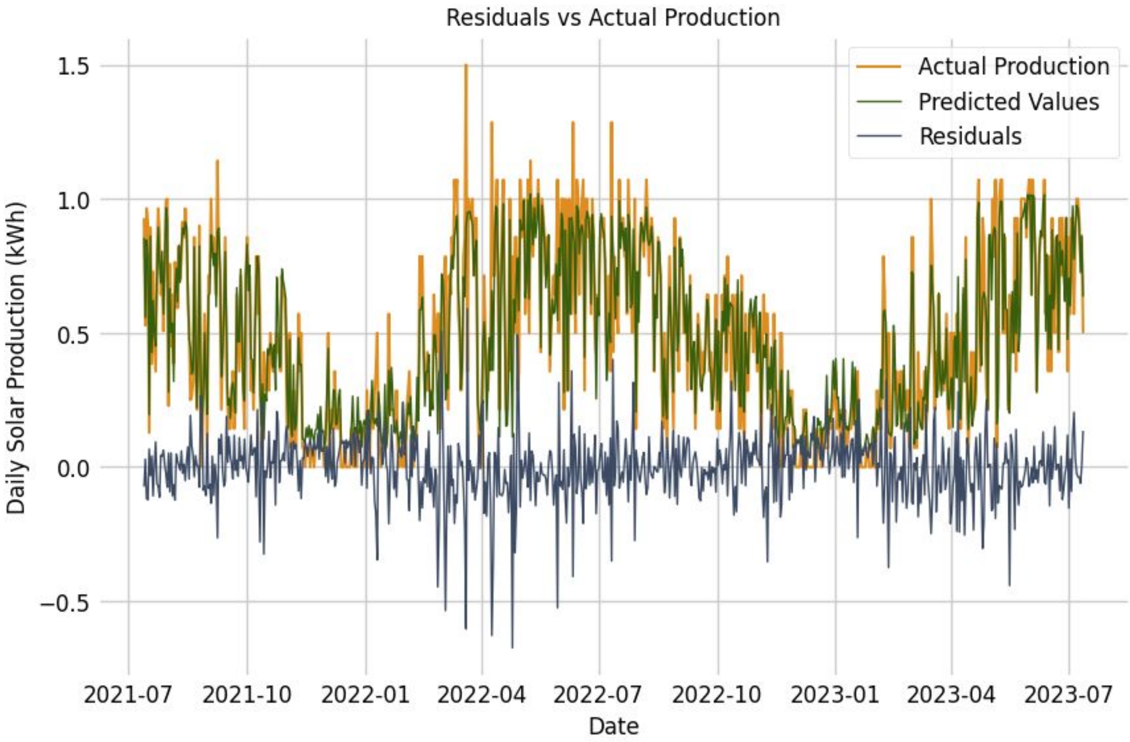
- 7 weather factors
- panel age

R^2 Score: 89%

Model and Forecasting



Model and Forecasting

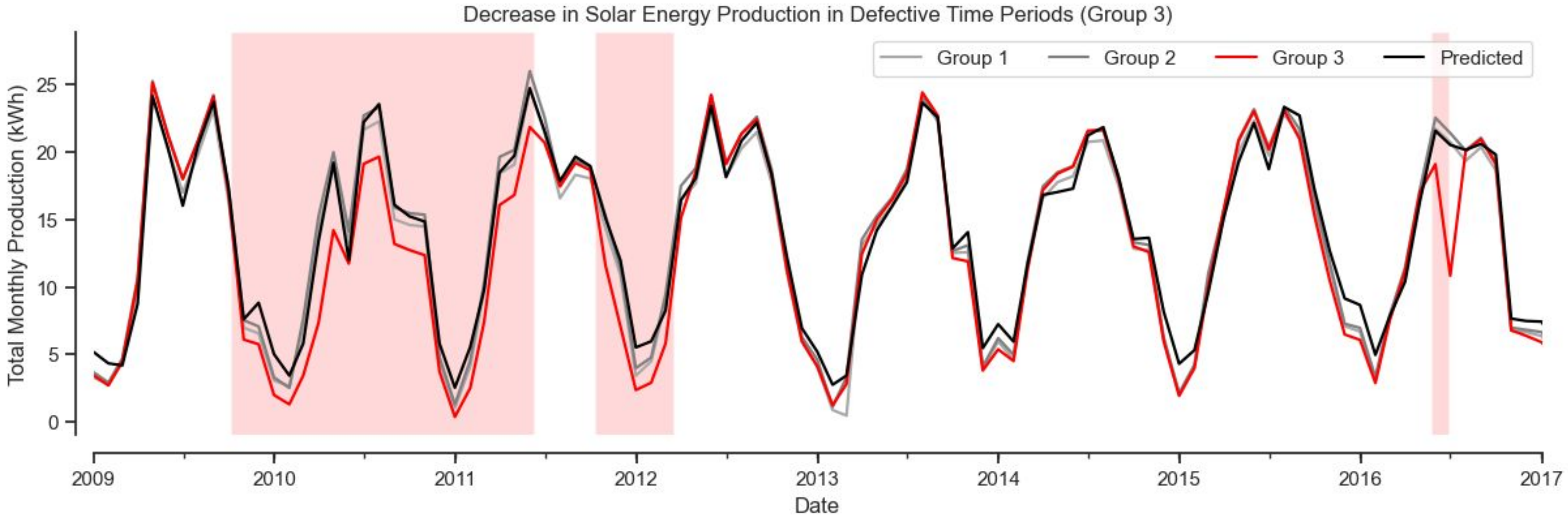


Challenges:

- Reduced performance
- Spring
- Winter

Losses due to Defects

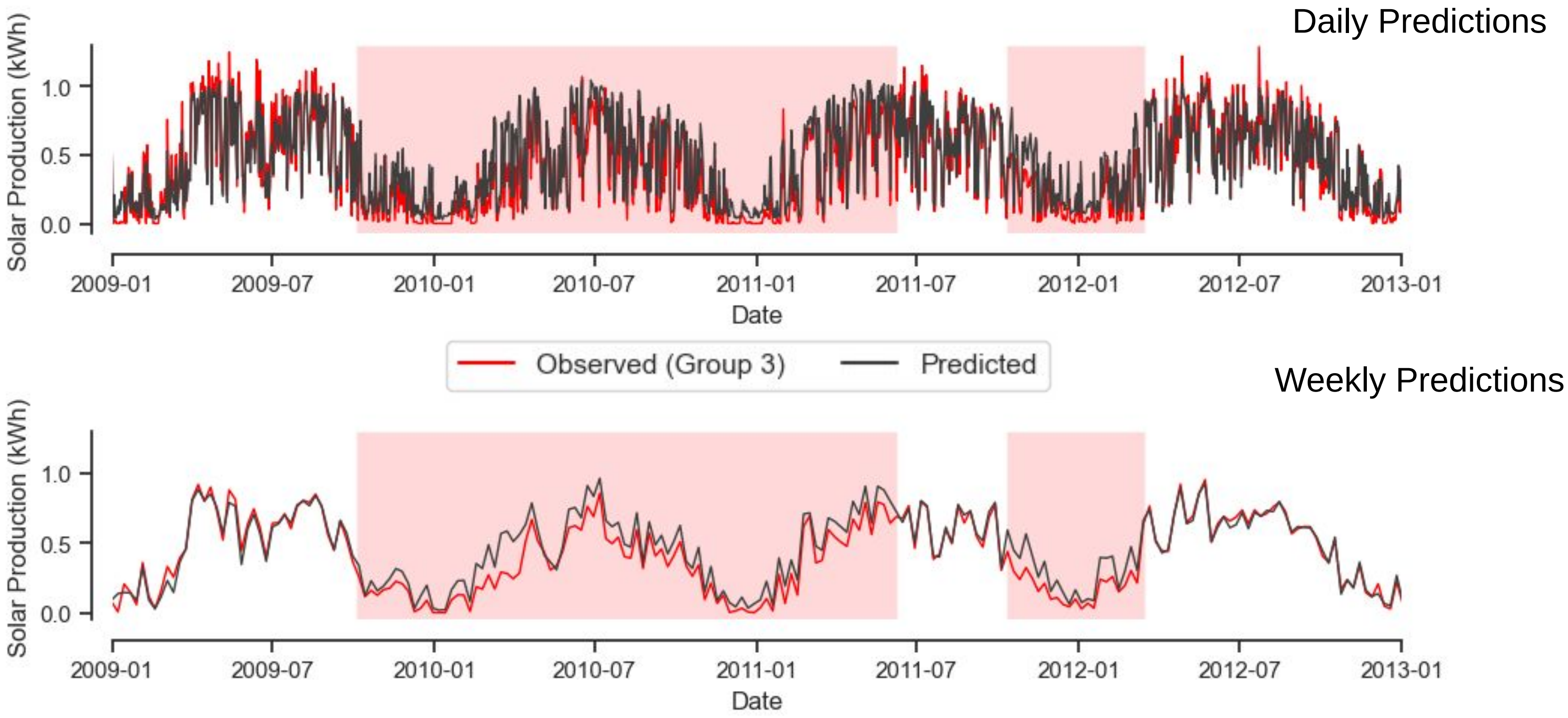
Comparing the Groups



Group 3 had multiple repairs - and issues undetected over a long time!

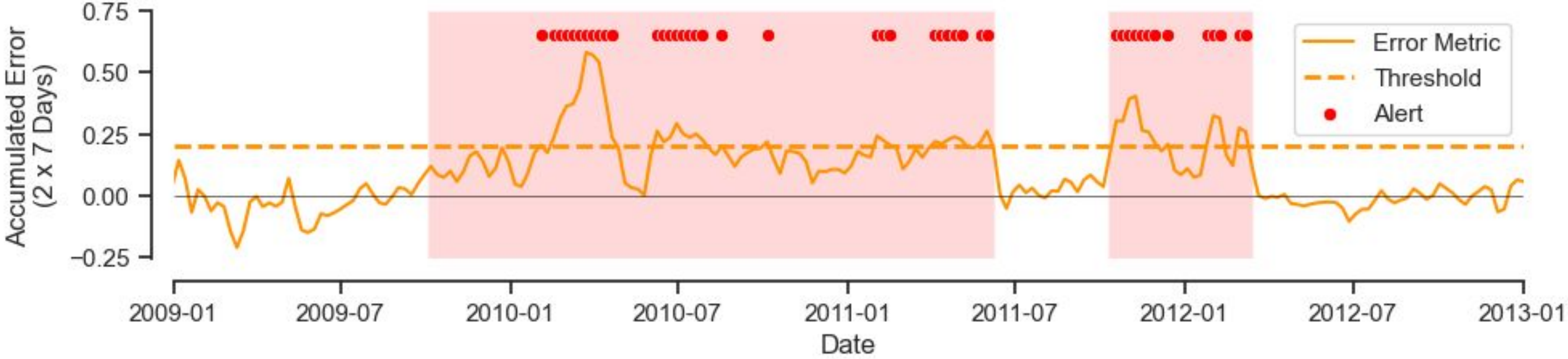
Defect Detection From Prediction

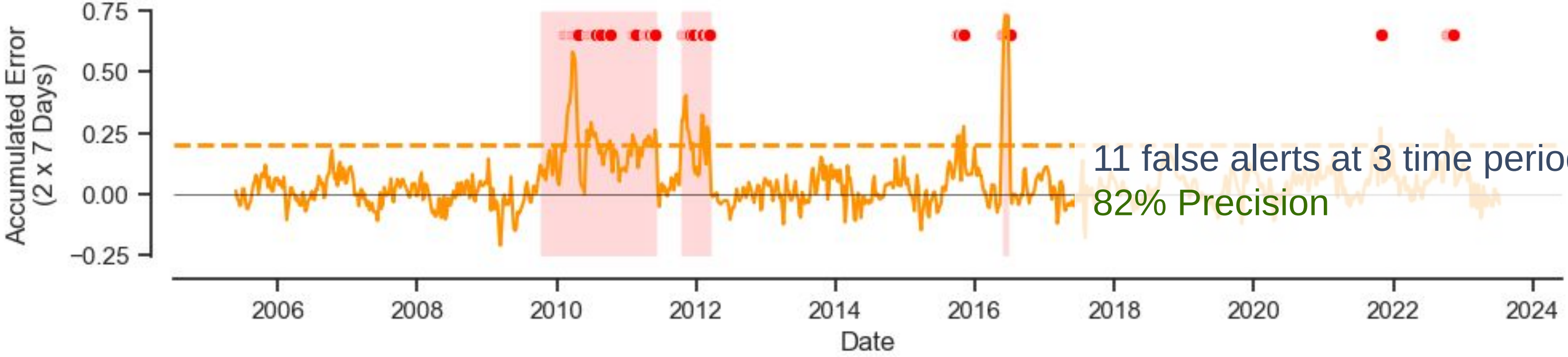
Daily Data varies a lot



Raising Alerts

Consider Accumulated Errors





11 false alerts at 3 time periods

82% Precision

Summary

- **Continuous increase** in German solar energy production with strong seasonality
- **Solar capacity** and **sunshine duration** are main factors for modeling Germany’s solar energy generation
- **Weather data** are good predictors for both national and local solar production
- **Panel age** does **not** influence the private solar energy production
- Modelling private solar production is **more noisy in winter**
- We can identify the **need for maintenance** using our predictions

Future: General model applicable to other households

Thank you for your attention

APPENDIX – Technical Details

Outlook

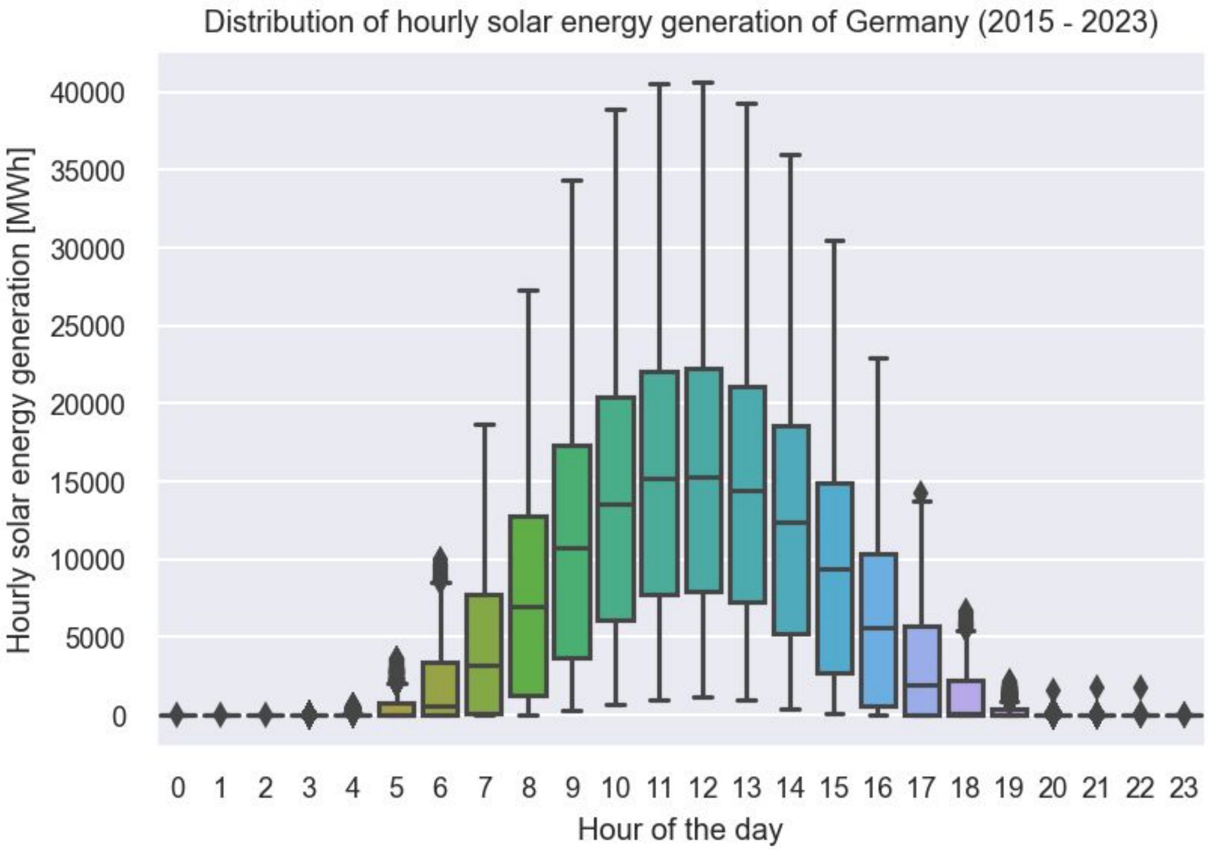
- Improve household predictions in winter

- Create a generalizable model for households:
    - Predict on daily change rather than daily production

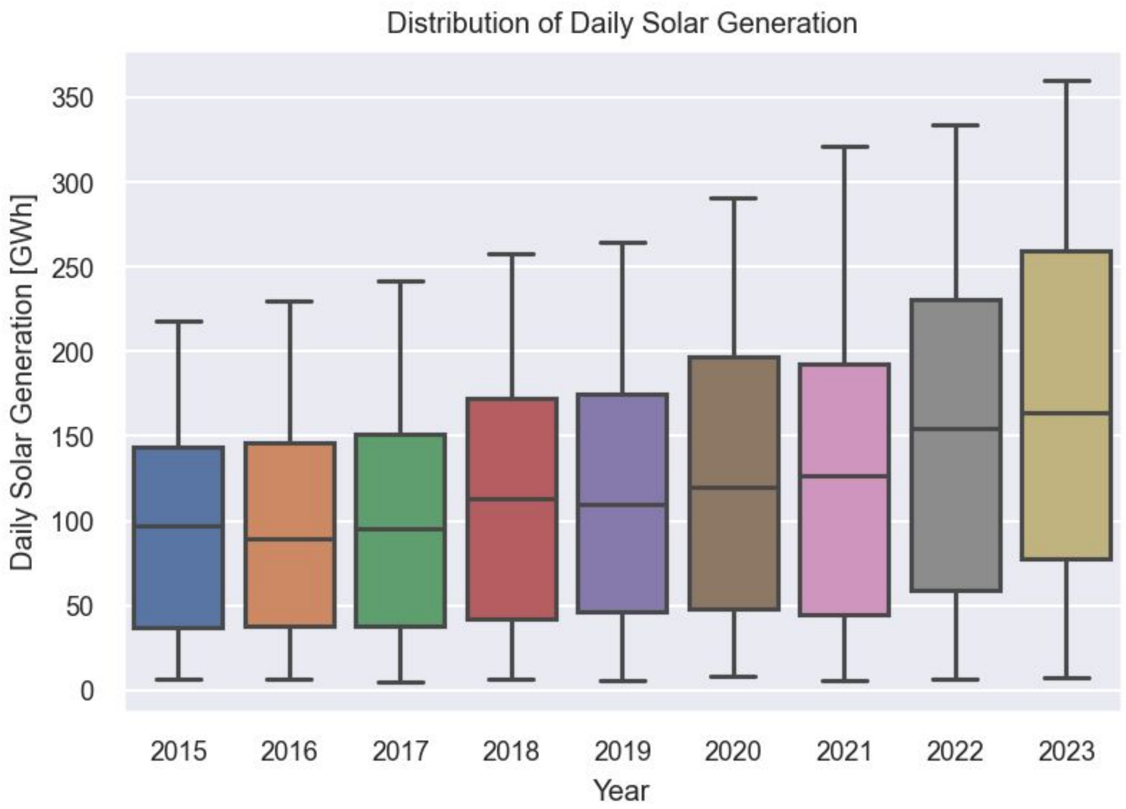
- Use of APIs to retrieve current data automatically
    - PV solar household data: https://api.openmeter.de/v1/docs#/
    - DWD weather data: https://dwd.api.bund.dev/

APPENDIX - National Solar Data

Hourly solar energy generation vs. hour of the day

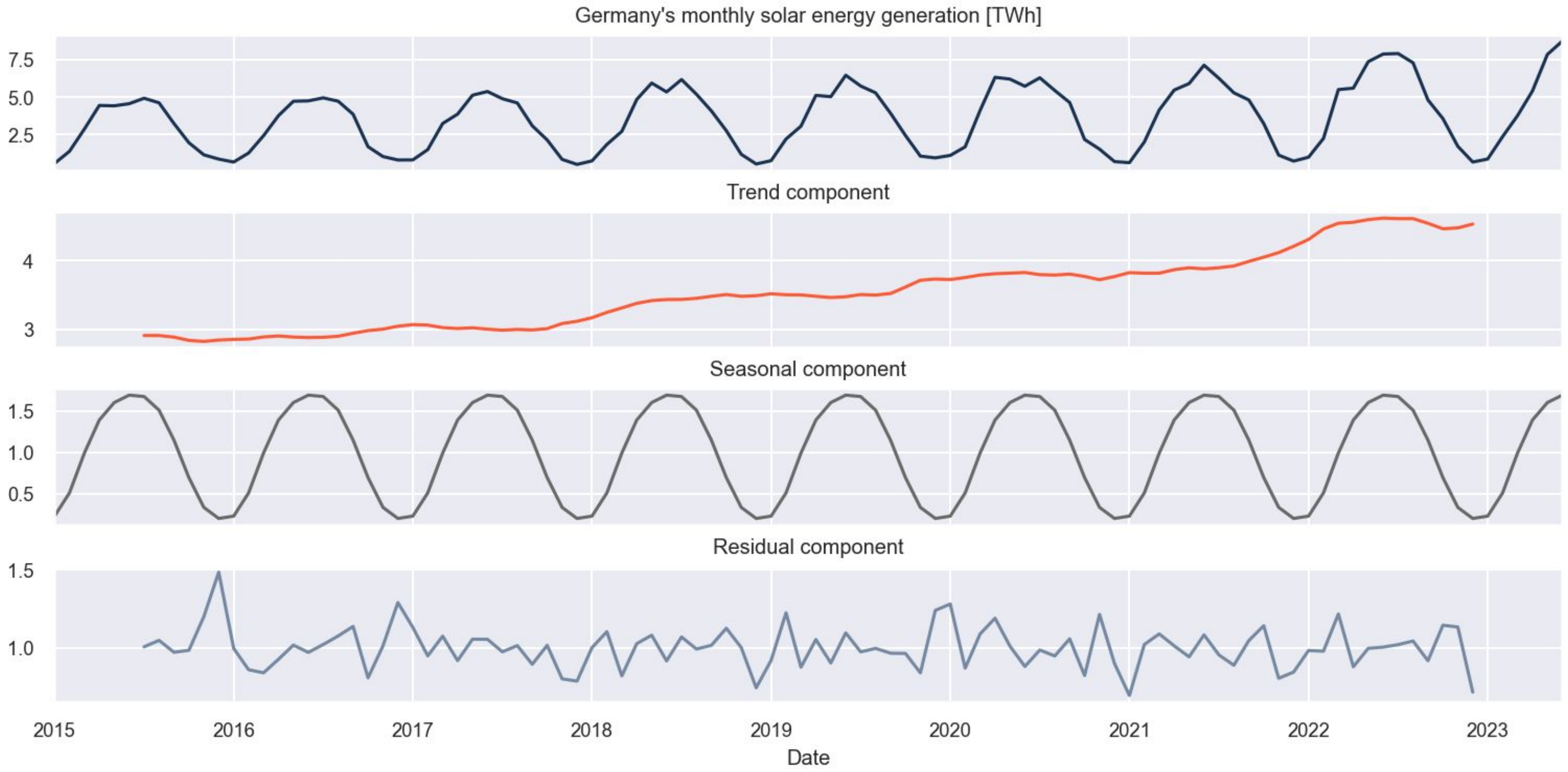


Daily solar energy generation vs. year



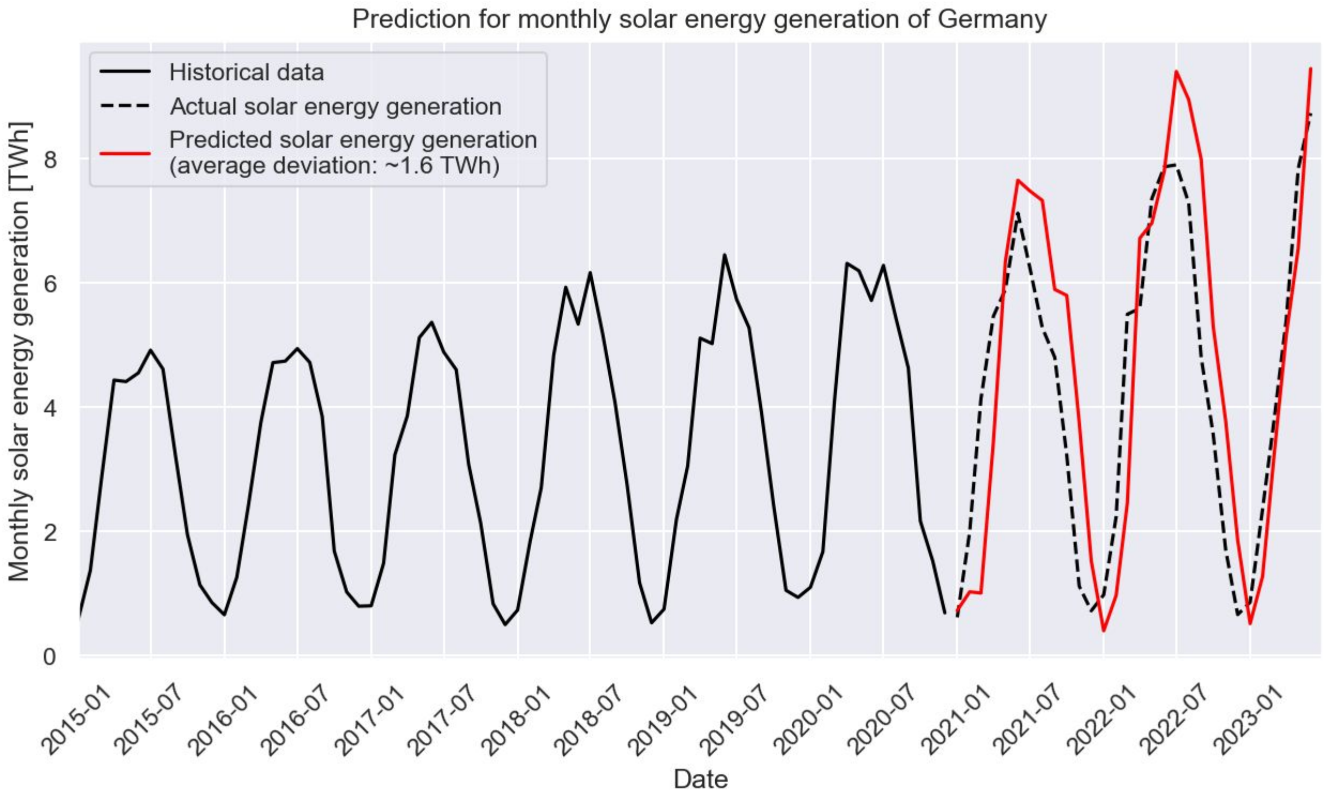
Monthly Solar Energy Generation

Time series decomposition: the basic idea

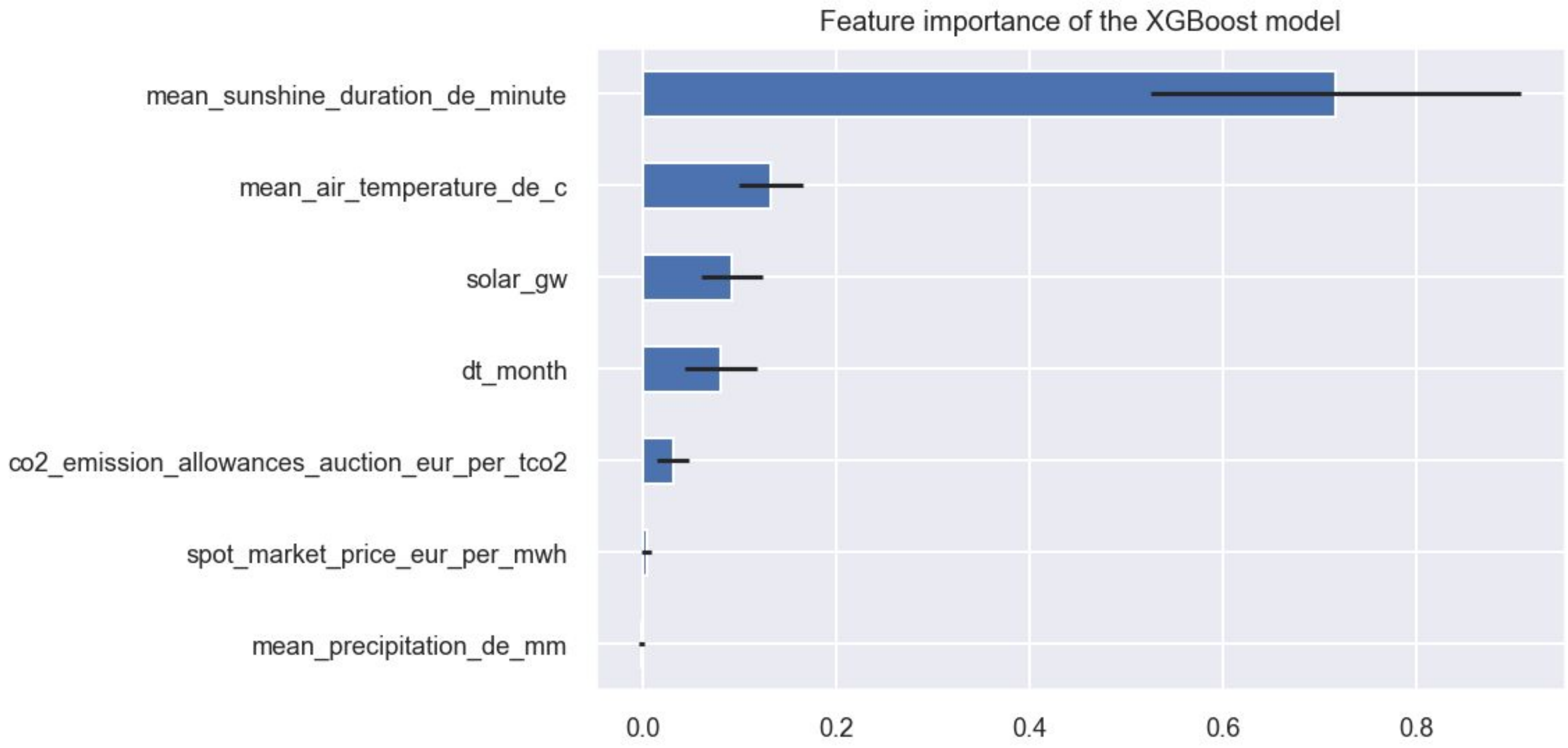


Solar Energy - Germany

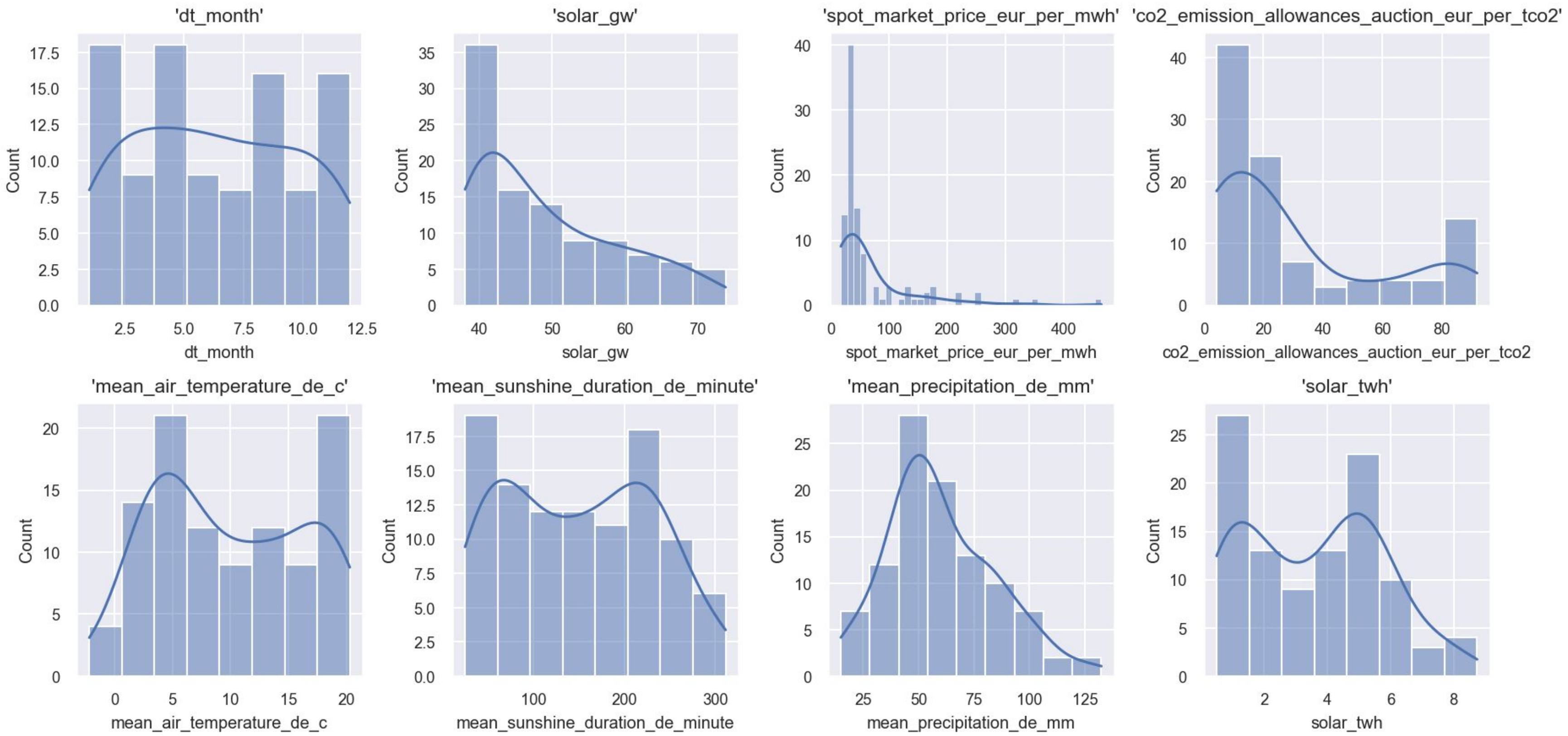
SARIMA-based time series forecasting



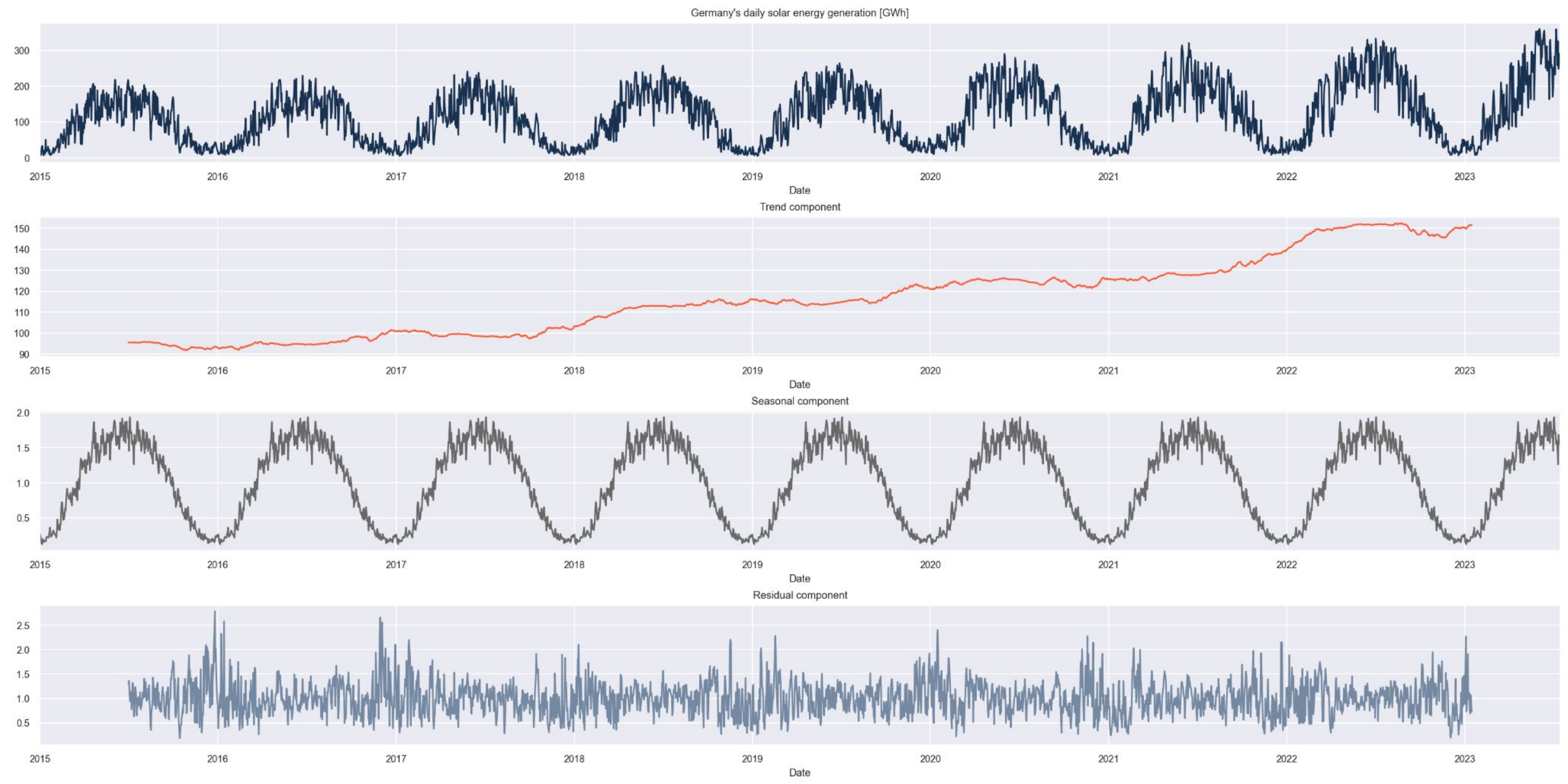
Feature importance of XGBoost model for Germany's monthly solar generation



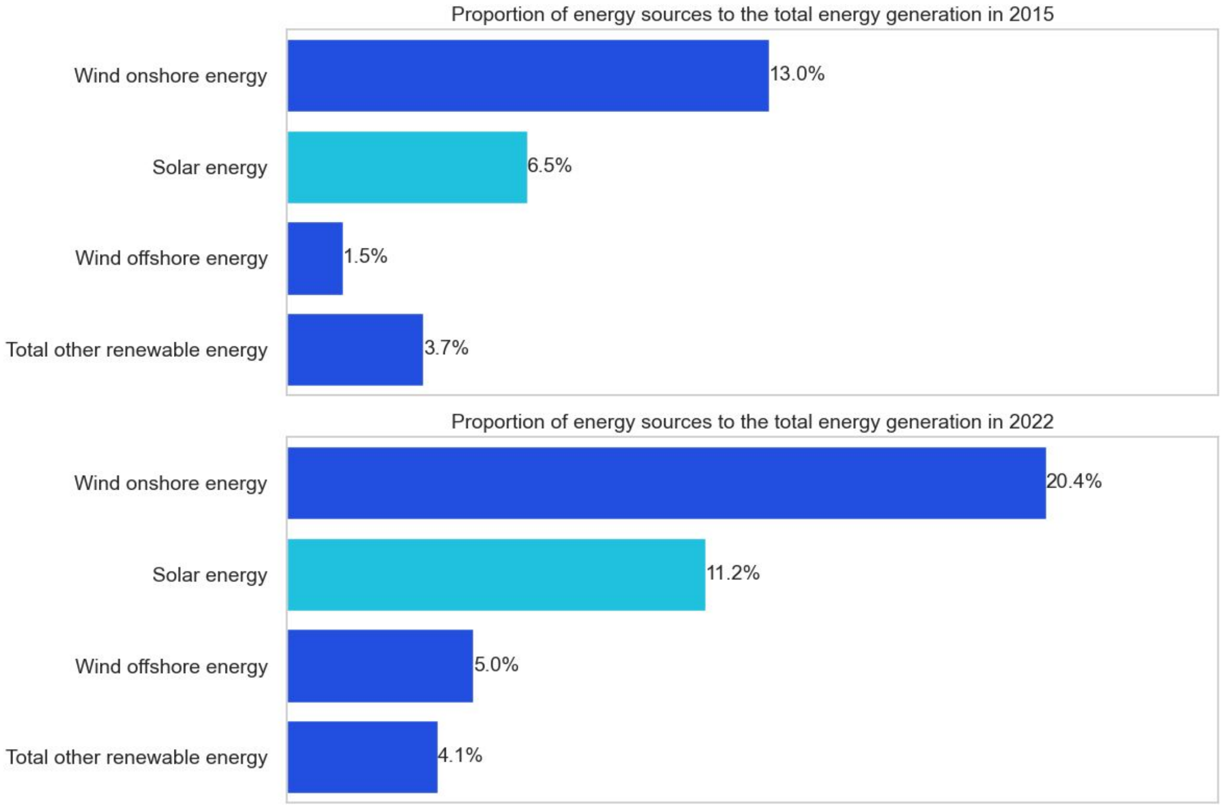
Distribution of features for modeling of Germany's monthly solar generation



Decomposition of daily solar energy generation



Energy Generation - Germany



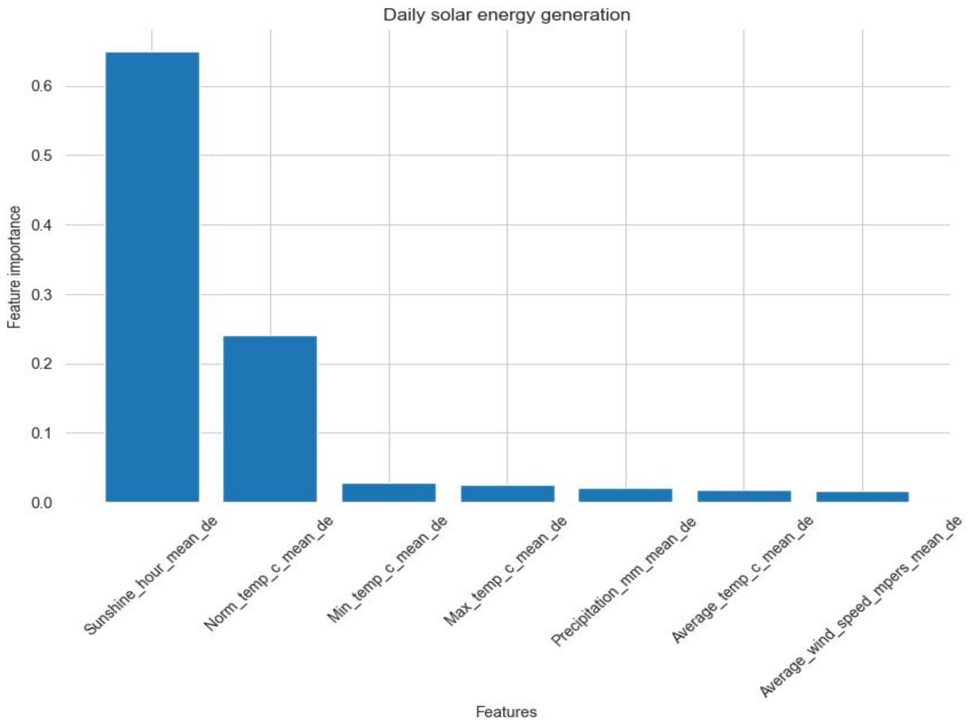
Solar Energy Generation - Germany

Daily solar energy production in Germany

Models	R2 Score from univariate analysis	R2 Score from multivariate analysis
XGBoost	0.6758	0.8884 (7 weather features) 0.7985 (Sun shine feature)
Random Forest	0.6172	0.8794 (7 weather features) 0.7446 (Sun shine feature)
LSTM	0.8494	0.9233 (7 weather features) 0.8528 (Sun shine feature)
Prophet		

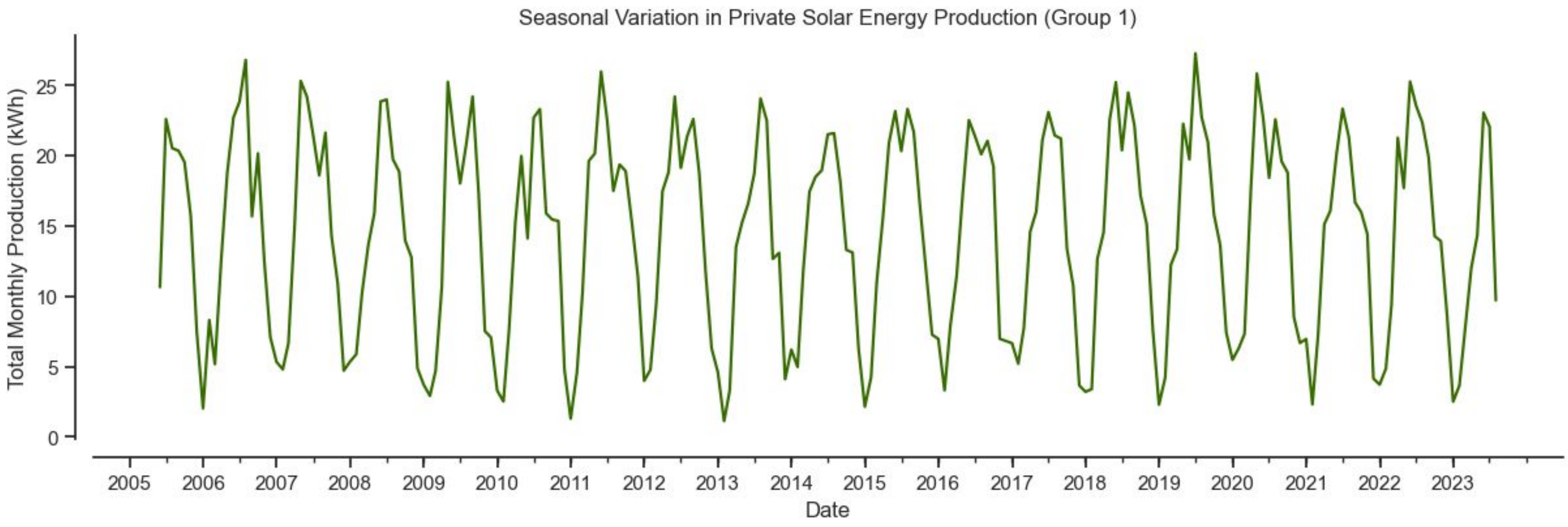
Solar Energy Generation – Germany

Weather features importance from XGBoost

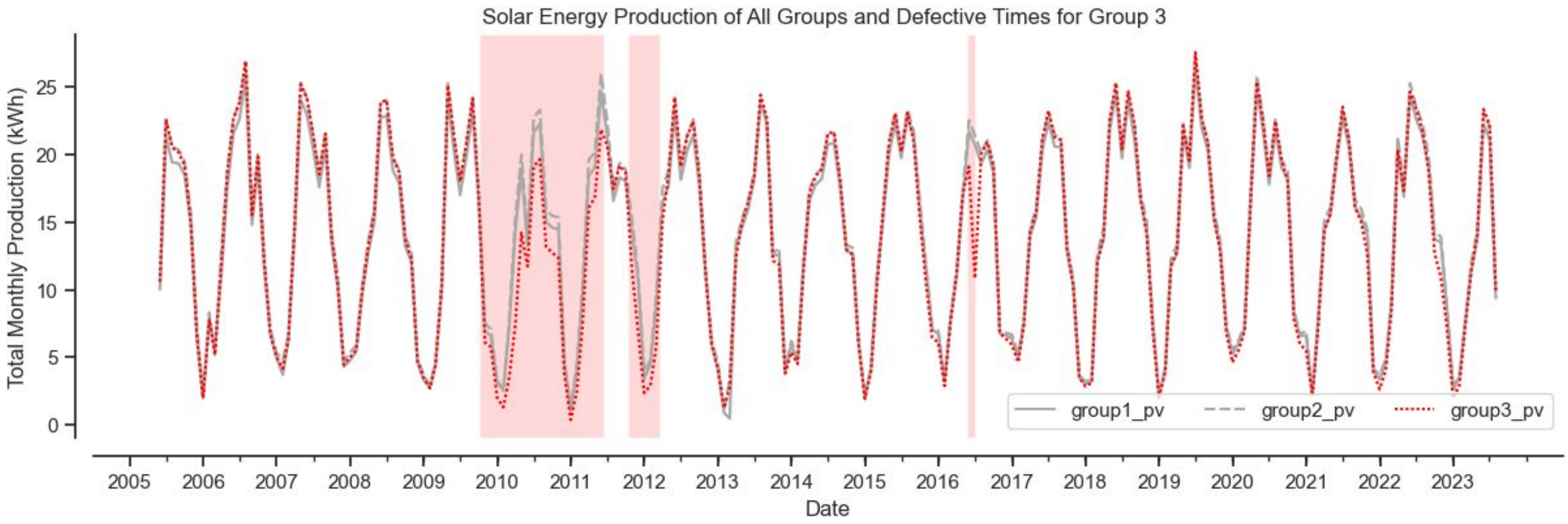


APPENDIX - Household GR Data

Energy Production - Household



Losses due to Defects



Group 3 had multiple repairs - and issues undetected over a long time!

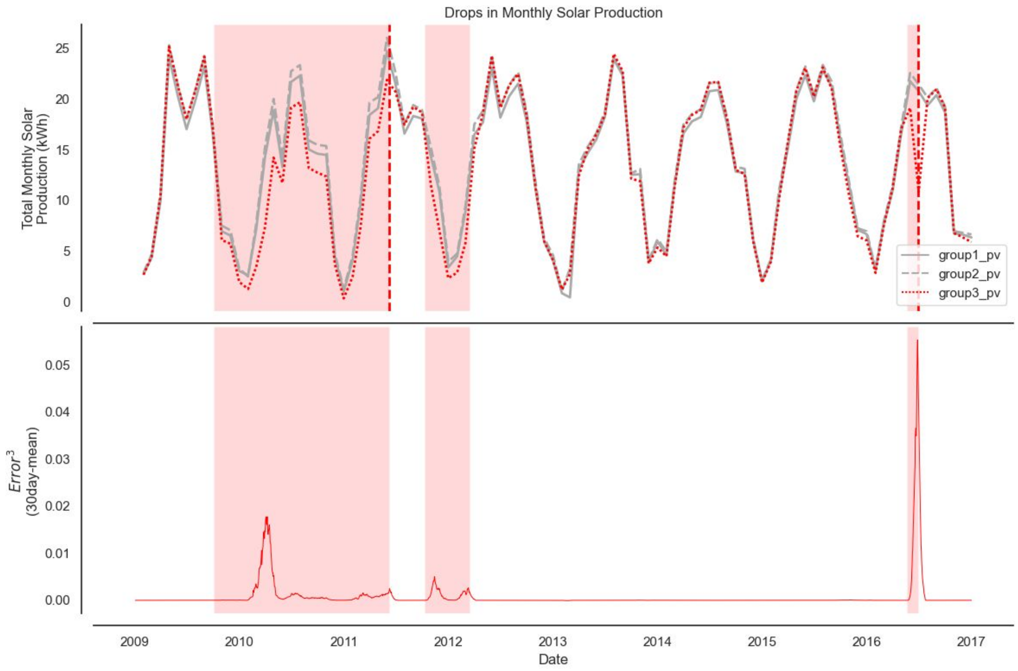
Semi-Manual Labelling Defects

Use production of other parties as reference for expected production

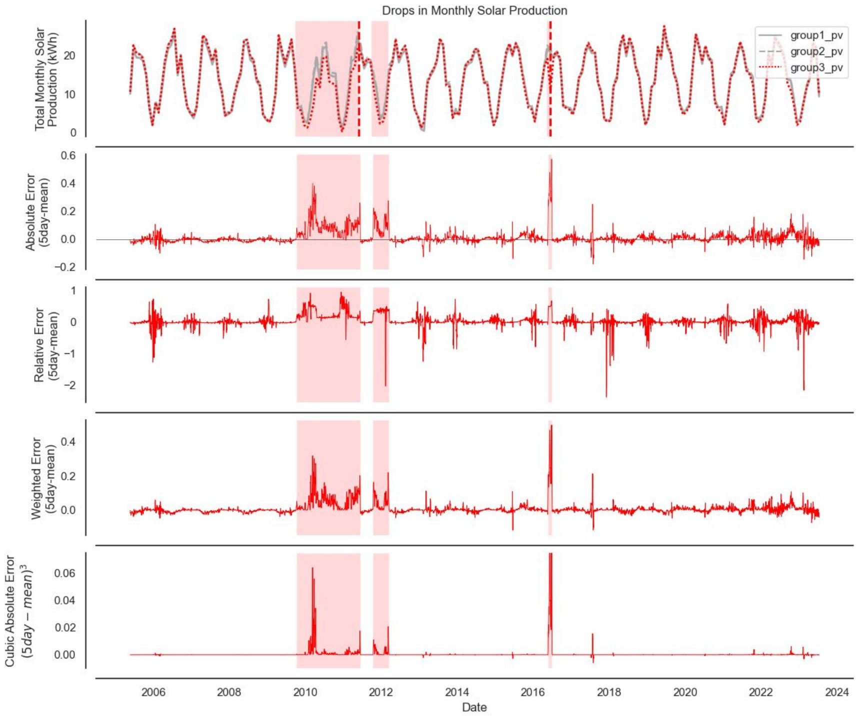
- Calculate expected daily mean
- Calculate absolute error (actual - expected)
- Calculate relative error (actual - expected)/expected

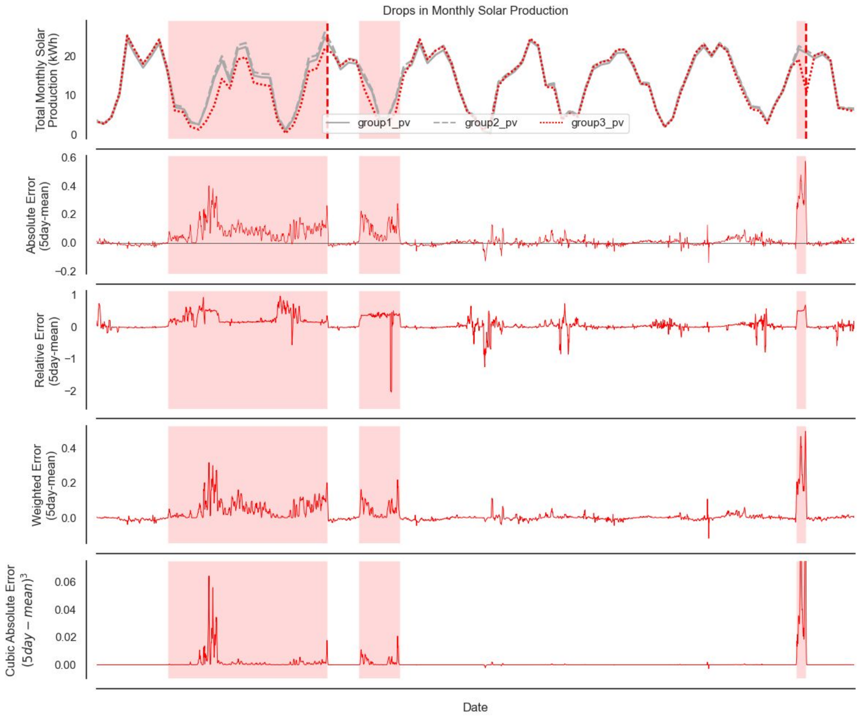
Semi-Automated labelling based on pre-calculated date ranges done

- Account for lag in error detection
- Account for difficulty to detect errors in winter



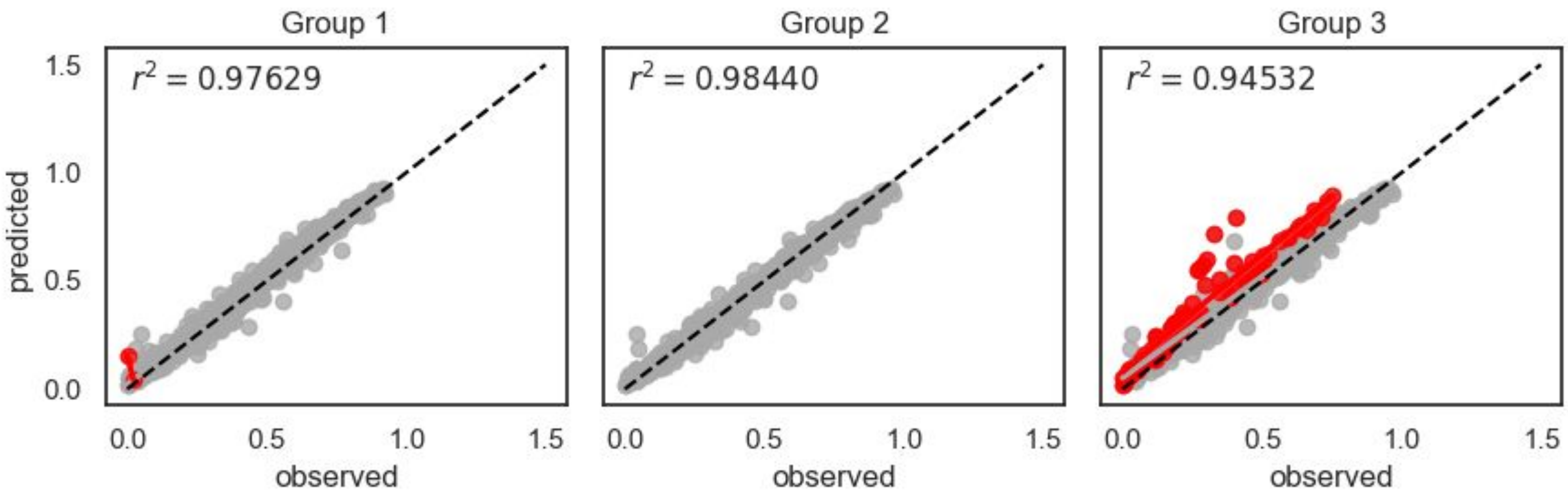
Semi-Manual Labelling Defects



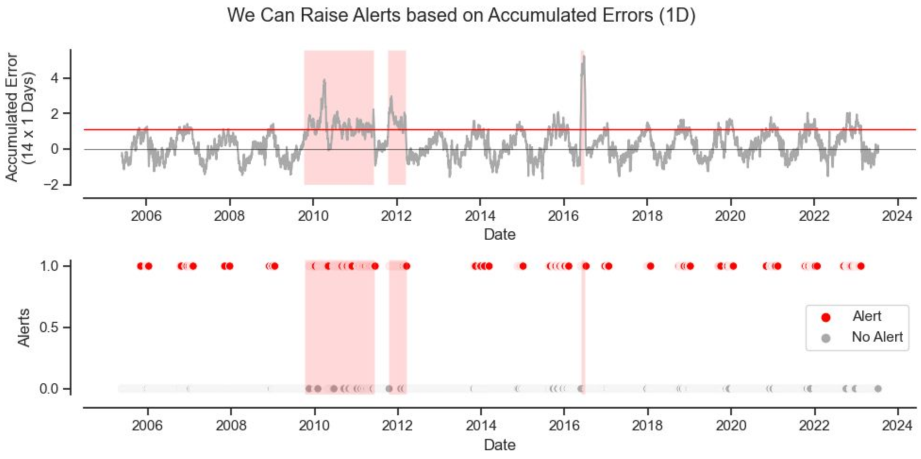


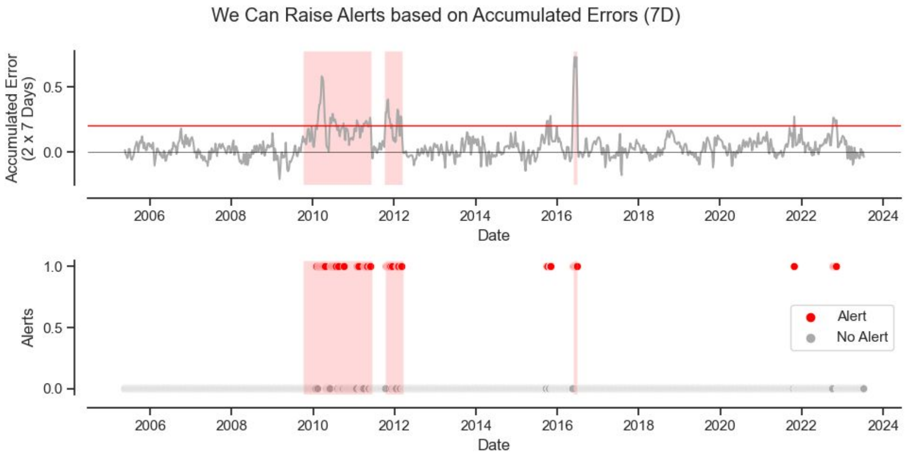
Residuals increase due to Defects

3d aggregated prior to prediction

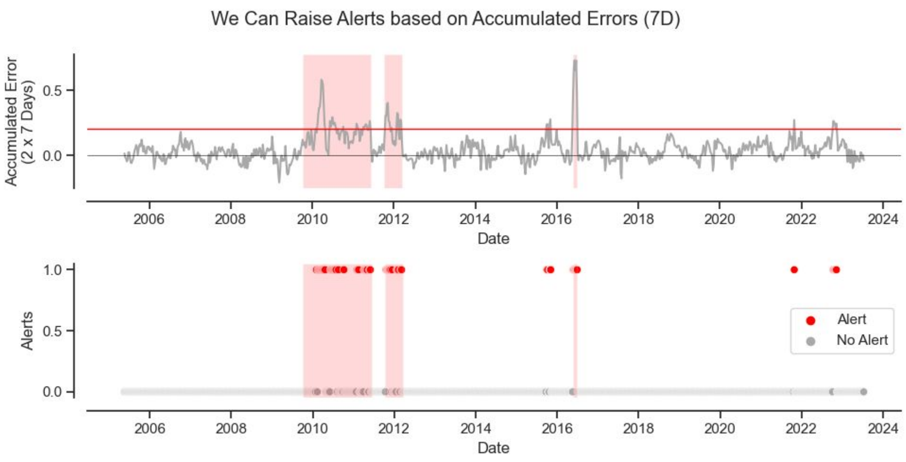


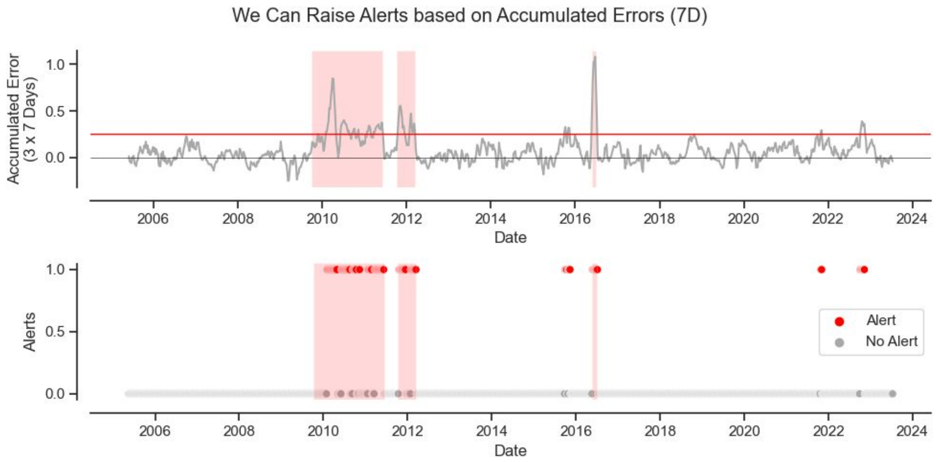
Smoothing Errors by Aggregation





Longer Time Window For Alerts





49 True Alerts
11 False Alerts in 3 Time Periods
67 Missing Alerts

(Threshold 0.2)

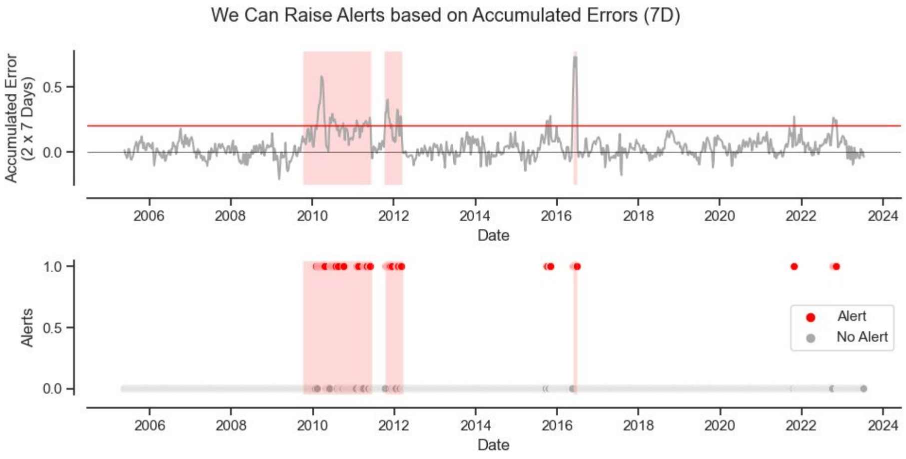
```
Recall: 0.42241
Precision: 0.81667
Accuracy: 0.91772
```

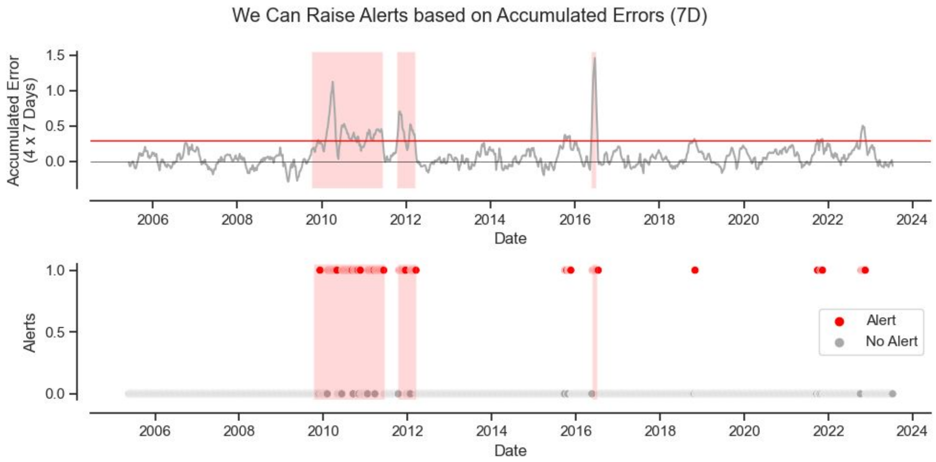
68 True Alerts
16 False Alerts in 3 Time Periods
48 Missing Alerts

(Threshold 0.25)

```
Recall: 0.58621
Precision: 0.80952
Accuracy: 0.93249
```

Longer Time Window For Alerts





49 True Alerts
11 False Alerts in 3 Time Periods
67 Missing Alerts

(Threshold 0.2)

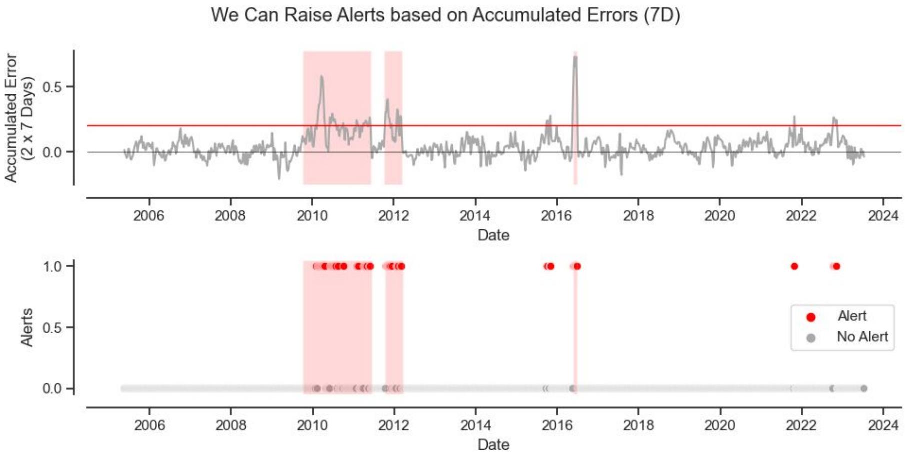
```
Recall: 0.42241
Precision: 0.81667
Accuracy: 0.91772
```

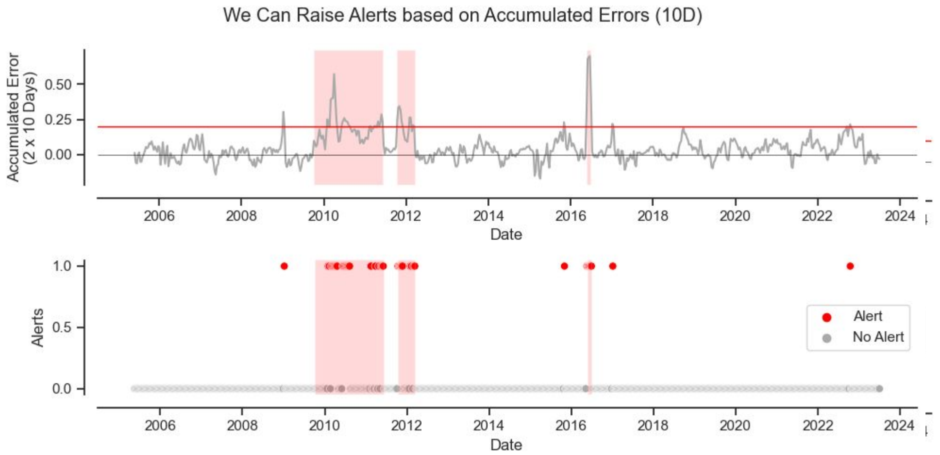
74 True Alerts
20 False Alerts in 4 Time Periods
42 Missing Alerts

(Threshold 0.2)

```
Recall: 0.63793
Precision: 0.78723
Accuracy: 0.93460
```

Longer Time Window For Alerts





49 True Alerts
11 False Alerts in 3 Time Periods
67 Missing Alerts

(Threshold 0.2)

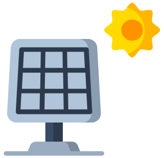
```
Recall: 0.42241
Precision: 0.81667
Accuracy: 0.91772
```

33 True Alerts
6 False Alerts in 2 Time Periods
48 Missing Alerts

(Threshold 0.2)

```
Recall: 0.40741
Precision: 0.84615
Accuracy: 0.91855
```

Methodological Strategies



Classification

- Label observations when defects occurred as target
- Include weather and energy production as features

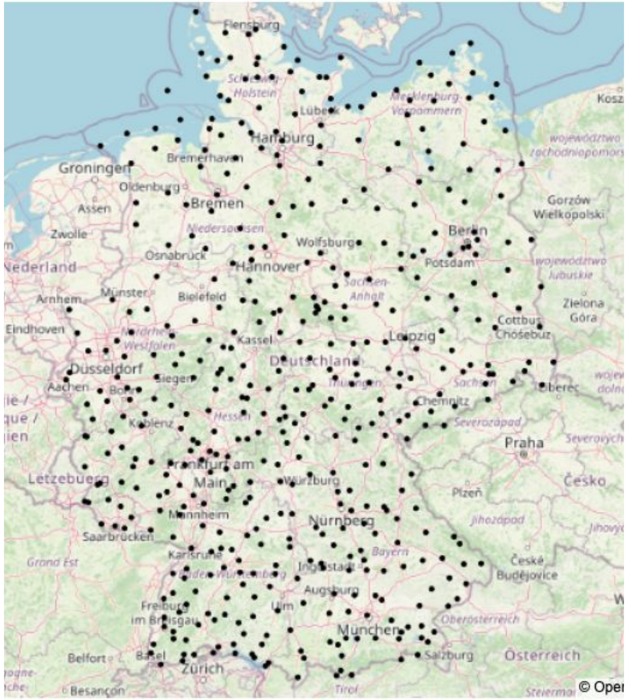
Regression

- Predict solar production
- Include weather information as features
- Detect deviation between observed and expected energy production

APPENDIX – Weather & Solar Data

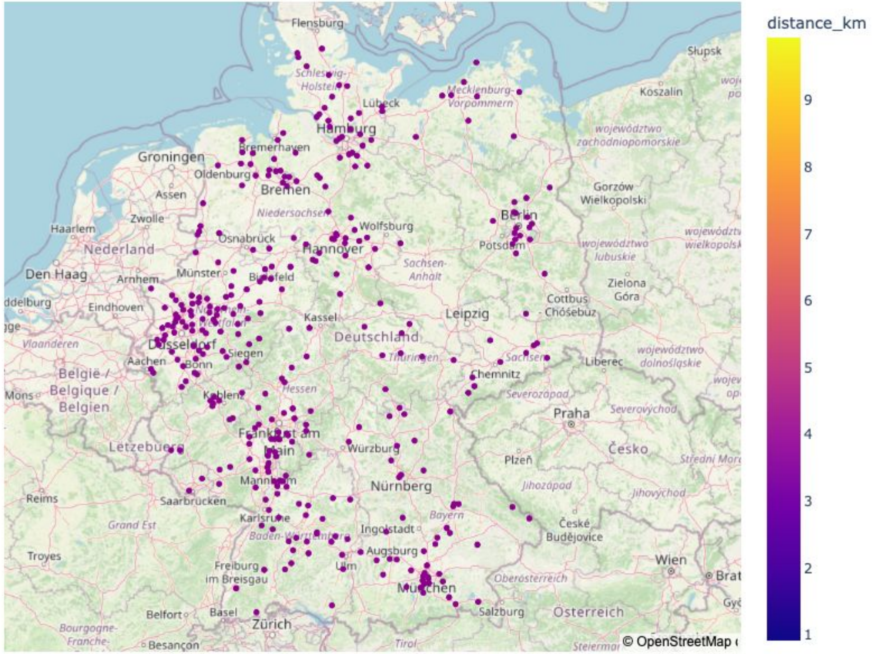
Weather Data

- Data by Deutscher Wetterdienst (DWD)
- Daily: 478 stations measured since 2005
- Hourly: 21 stations measured since 2005



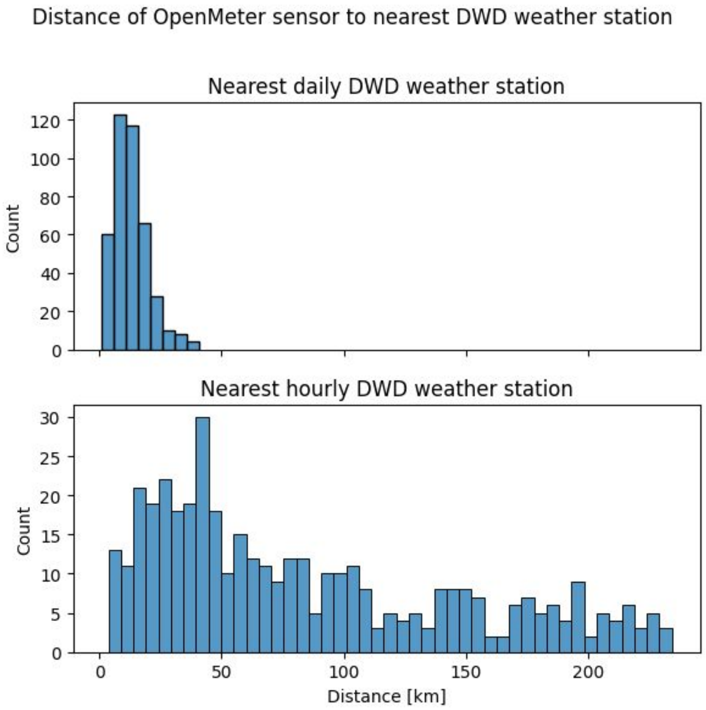
Available Solar Data - Households

- 412 datasets on private energy production available at OpenMeter platform (https://www.openmeter.de/)
- Data since 2018 (or later)
- 15min resolution



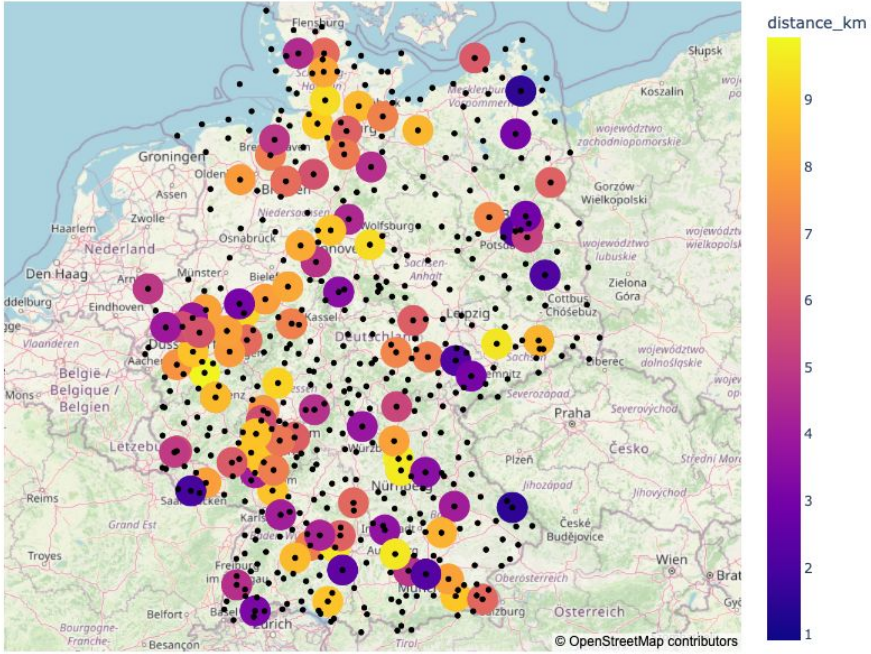
Available Data - Solar Households

- 412 datasets on private energy production available at OpenMeter platform (https://www.openmeter.de/)
- Data since 2018 (or later)
- 15min resolution
- nearby DWD weather station with daily weather (or hourly weather)



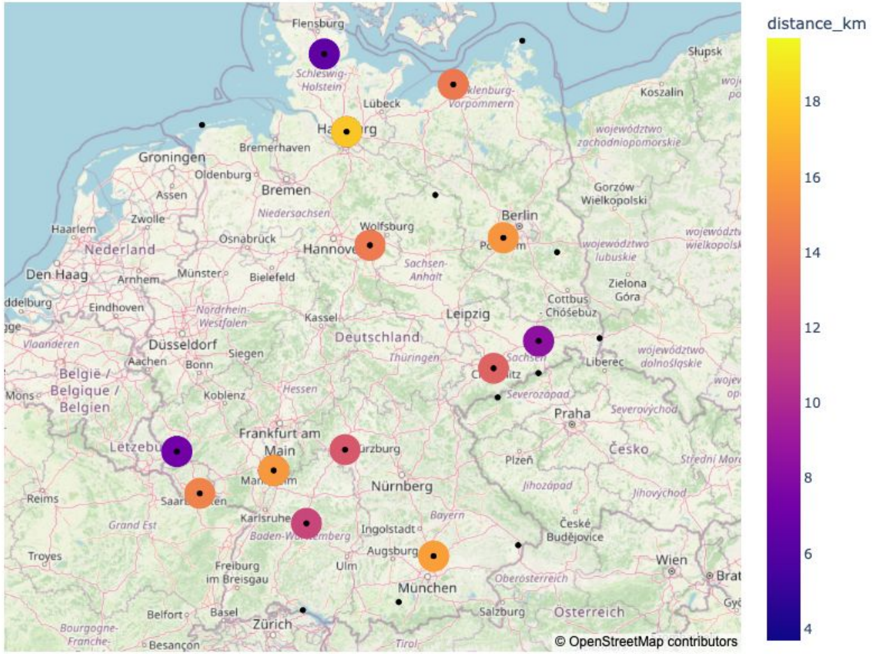
Available Data – Solar Households

- 412 datasets on private energy production available at OpenMeter platform (https://www.openmeter.de/)
- Data since 2018 (or later)
- 15min resolution
- nearby DWD weather station
    - 164 OM sensors with <10 km distance to daily DWD station



Available Data – Solar Households

- 412 datasets on private energy production available at OpenMeter platform (https://www.openmeter.de/)
- Data since 2018 (or later)
- 15min resolution
- nearby DWD weather station
    - 164 OM sensors with <10 km distance to daily DWD station
    - 16 OM sensors with <10km distance to DWD station
- Sometimes even near hourly weather station



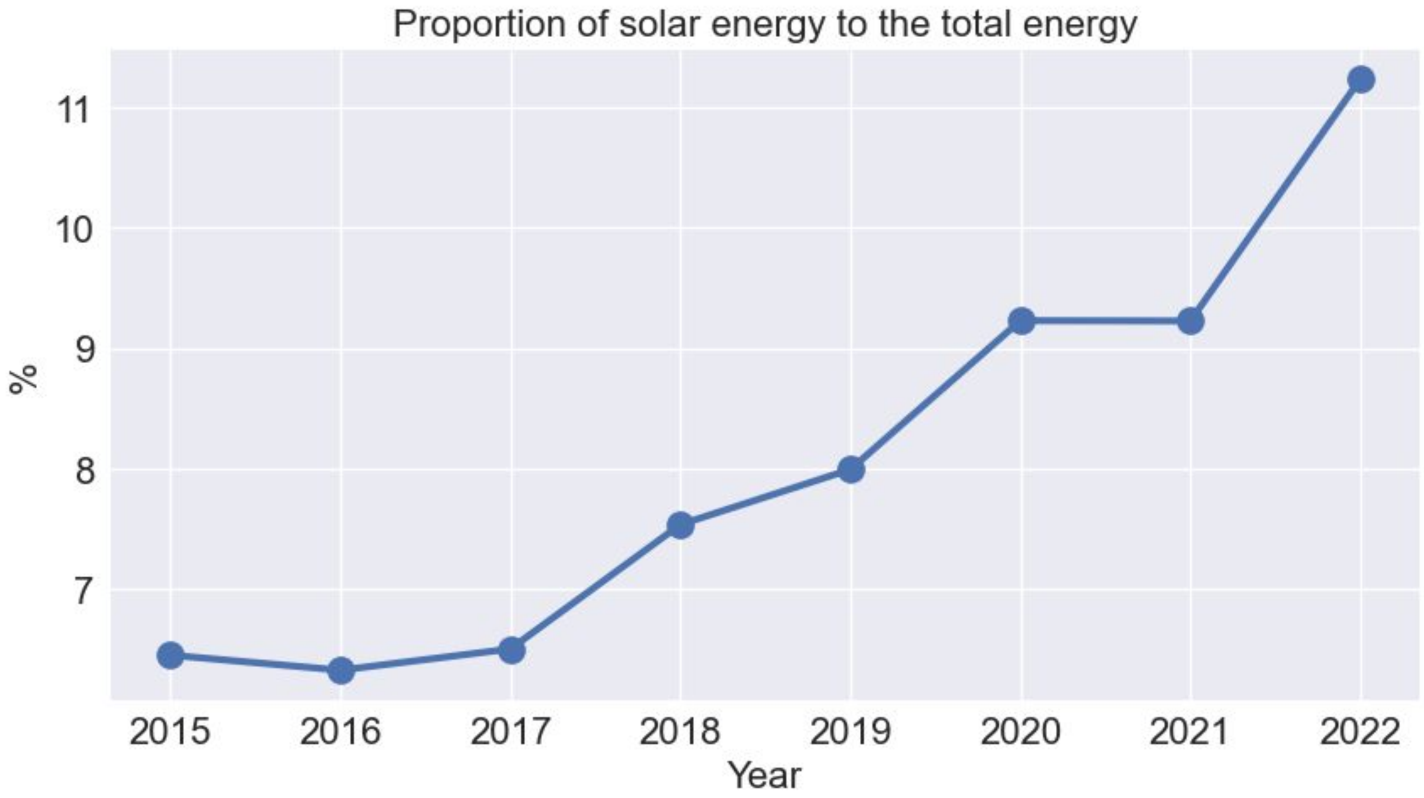
Future Work

Classification model to detect defective solar panels

Use of APIs to retrieve current data

- PV solar household data: https://api.openmeter.de/v1/docs#/
- DWD weather data: https://dwd.api.bund.dev/

Solar Energy Generation - Germany



ICONS

Our Team

SX

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Semiconductor Technologist

JS

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Assembling/Welding Technologist

BD

Bimla Danu

Theoretical Condensed Matter Physicist

KS

Kathrin Seibt

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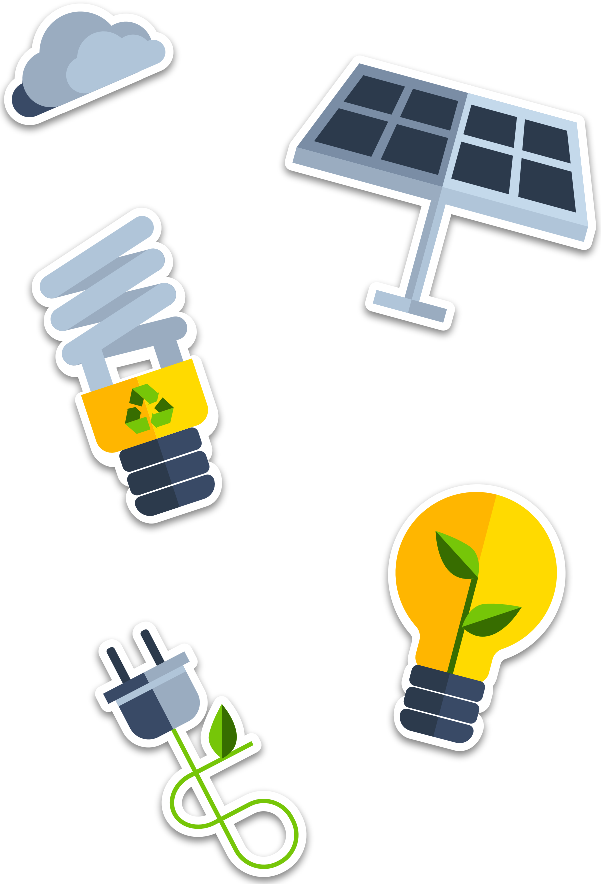
Kathrin Seibt

Natural Scientist / Bioinformatician

“The amount of solar energy hitting the earth in one hour is more than enough to power the world for one year.”

—University of California, DAVIS

ORIGINAL TEMPLATE SLIDES



SOLAR ENERGY COMPANY PITCH DECK

Here is where your presentation begins

CONTENTS OF THIS TEMPLATE

This is a slide structure based on a pitch deck presentation
You can delete this slide when you’re done editing the presentation

FONTS	To view this template correctly in PowerPoint, download and install the fonts we used
USED AND ALTERNATIVE RESOURCES	An assortment of graphic resources that are suitable for use in this presentation
THANKS SLIDE	You must keep it so that proper credits for our design are given
COLORS	All the colors used in this presentation
INFOGRAPHIC RESOURCES	These can be used in the template, and their size and color can be edited
CUSTOMIZABLE ICONS	They are sorted by theme so you can use them in all kinds of presentations

For more info:
SLIDESGO | SLIDESGO SCHOOL | FAQS

You can visit our sister projects:
FREEPIK | FLATICON | STORYSET | WEPIK | VIDFY

TABLE OF CONTENTS

01

PROBLEM VS. SOLUTION

You can describe the topic of the section here

INTRODUCTION

Mercury is the closest planet to the Sun and the smallest one in the Solar System—it's only a bit larger than the Moon. The planet's name has nothing to do with the liquid metal, since Mercury was named after the Roman messenger god

OUR COMPANY

Mercury is the closest planet to the Sun and the smallest one in the Solar System—it's only a bit larger than the Moon. The planet's name has nothing to do with the liquid metal

OUR TEAM

JENNA DOE

You can replace the image on the screen with your own

TOMMY LEAN

You can replace the image on the screen with your own

SUSAN BONES

You can replace the image on the screen with your own

PROBLEM

Do you know what helps you make your point clear?
Lists like this one:

- They're simple
- You can organize your ideas clearly
- You'll never forget to buy milk!

And the most important thing: the audience won't miss the point of your presentation

THEM

Mercury is the closest planet to the Sun and the smallest one in the Solar System—it's only a bit larger than the Moon

US

Venus has a beautiful name and is the second planet from the Sun. It's hot and has a poisonous atmosphere

320,000

Big numbers catch your audience’s attention

SOLUTION

Venus has a beautiful name and is the second planet from the Sun. It’s terribly hot—even hotter than Mercury—and its atmosphere is extremely poisonous

SWOT ANALYSIS

STRENGTHS	S	Mercury is the smallest planet of them all
WEAKNESSES	W	Venus has a beautiful name, but also high temperatures
OPPORTUNITIES	O	Earth is the beautiful planet where we all live on
THREATS	T	Despite being red, Mars is actually a cold place

PRODUCT OVERVIEW

MERCURY

Mercury is the smallest planet of them all

VENUS

Venus is the second planet from the Sun

EARTH

Earth is the only planet known to harbor life

MARS

Despite being red, Mars is actually a cold place

SATURN

Saturn is not the only planet with rings

OUR PLANS

BASIC

Mercury is the closest planet to the Sun and the smallest of them all

$10,000

PRO

Venus has a beautiful name and is the second planet from the Sun

$19,000

PREMIUM

Despite being red, Mars is actually a cold place. It's full of iron oxide dust

$26,000

A PICTURE IS WORTH A THOUSAND WORDS

PRODUCT
DEMO

You can replace the image on the screen with your own work. Just right-click on it and select “Replace image”

TRACTION

$140,000

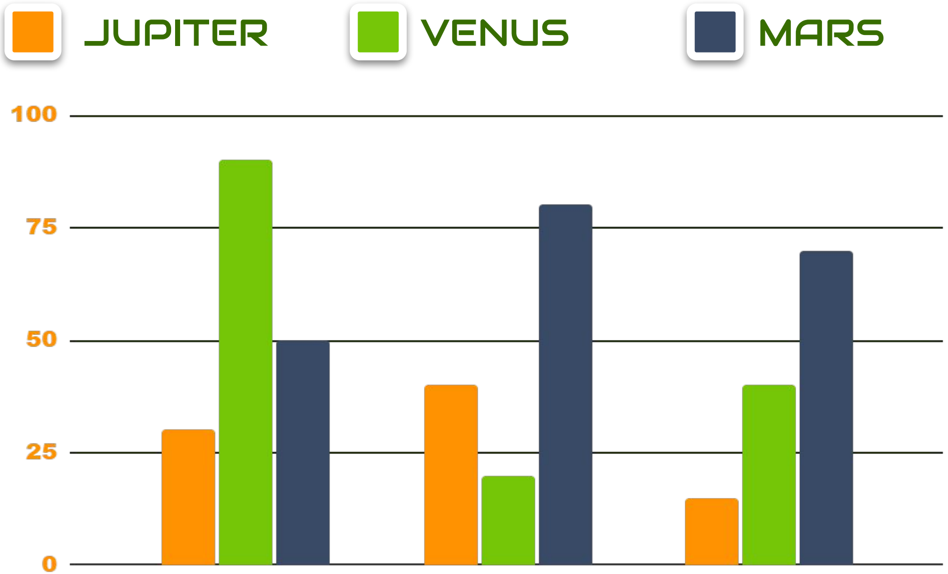
Gross revenue in 2020

12,000

Registered businesses

$150

Home energy savings



Follow the link in the graph to modify its data and then paste the new one here.
For more info, click here

AWESOME WORDS

CASE STUDY

CHALLENGE

Despite being red, Mars is actually a cold place full of iron oxide dust

BUDGET

Jupiter is a gas giant and the biggest planet in the entire Solar System

RESULTS

Venus has a beautiful name and is the second planet from the Sun

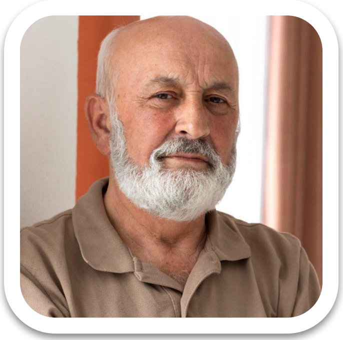
SOLUTIONS

Saturn is a gas giant, composed mostly of hydrogen and helium

REVIEWS

NICOLE WILLS

“Mercury is the closest planet to the Sun and the smallest one in the Solar System”



RON SMITH

“Jupiter is the biggest planet in the Solar System and the fourth-brightest object in the sky”

LATEST AWARDS

MERCURY

Mercury is the smallest planet of them all

JUPITER

Jupiter is the biggest planet of them all

NEPTUNE

Neptune is the farthest planet from the Sun

“This is a quote, words full of wisdom that someone important said and can make the reader get inspired.”

—SOMEONE FAMOUS

MARKET SIZE

60%

MERCURY

Mercury is the closest planet to the Sun

40%

VENUS

Venus is the second planet from the Sun

TARGET

AGE

10%	50%	40%
23 - 30	31 - 45	46+

SPEND PER CUSTOMER

$200

Venus has a beautiful name, but also high temperatures

GENDER

Man	30%
Woman	50%
Others	20%

INTERESTS

Devices

Home

Savings

Transportation

OUR COMPETITORS

	COMPETITOR A	COMPETITOR B	COMPETITOR C
DESCRIPTION	Mercury is the smallest planet of them all	Venus is terribly hot, even hotter than Mercury	Phobos and Deimos are both Mars’s moons
CHARACTERISTIC	Despite being red, Mars is actually a cold place	Jupiter is a gas giant planet in the Solar System	Saturn is a gas giant and has several rings
PRODUCT	Neptune is the farthest planet from the Sun	Earth is where we all live on and a blue planet	The Sun is the star at the center of the Solar System

A PICTURE ALWAYS REINFORCES THE CONCEPT

Images reveal large amounts of data, so remember: use an image instead of a long text. Your audience will appreciate it

BUSINESS MODEL

SATURN
Saturn is not the only planet with rings

EARTH
Earth is the only planet known to harbor life

JUPITER
Jupiter is the biggest planet of them all

VENUS
Venus is the second planet from the Sun

MARS
Despite being red, Mars is actually a cold place

MERCURY
Mercury is the smallest planet of them all

TIMING

WEEK 01	WEEK 02	WEEK 03	WEEK 04
MARS	**VENUS**	**SATURN**	**JUPITER**
Despite being red, Mars is cold	Venus has a beautiful name	Saturn is a gas giant with rings	Jupiter is a very big planet

PREDICTED GROWTH

80%

Increased income

$60M

Expected income

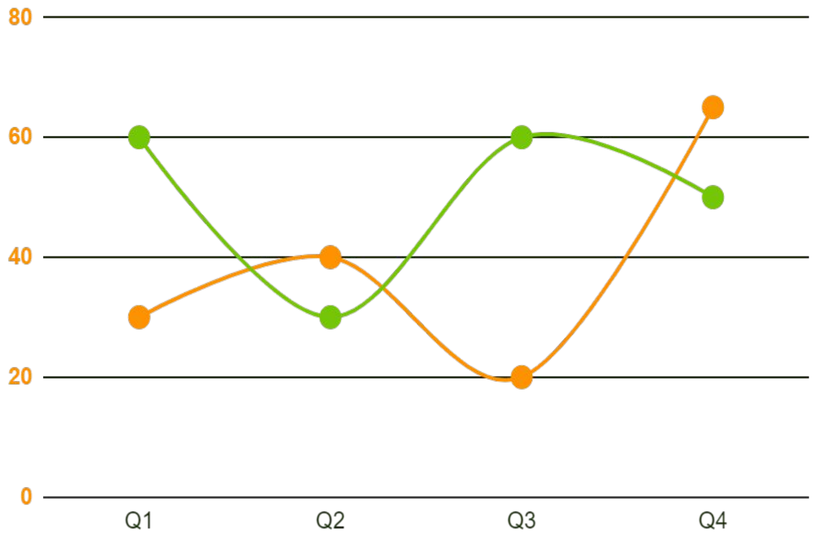
65%

More employees

2,000

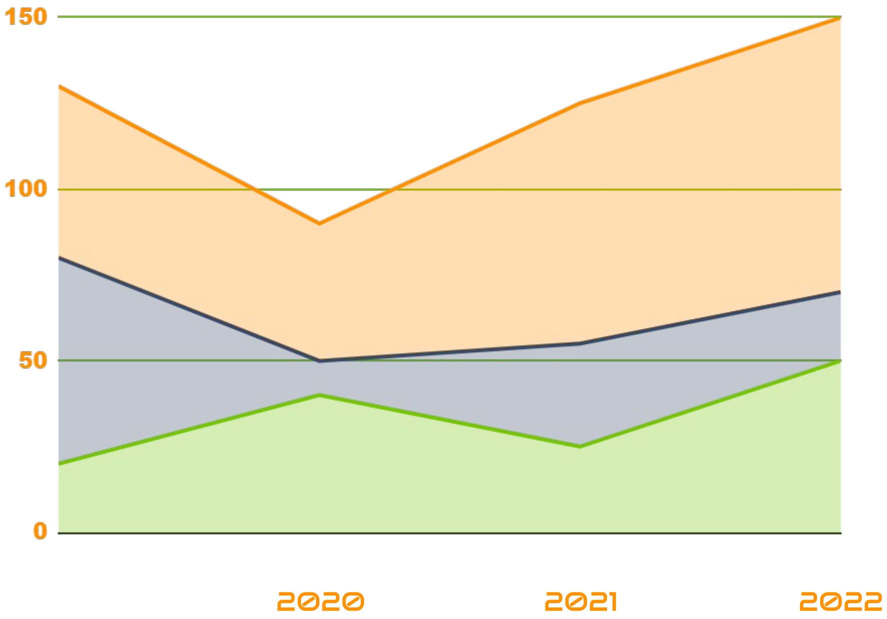
Total employees

Follow the link in the graph to modify its data and then paste the new one here. **For more info, click here**



INVESTMENT

20%	**VENUS** Venus is the second planet from the Sun
30%	**MARS** Mars is full of iron oxide dust
50%	**EARTH** Earth is the third planet from the Sun



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